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# LLMs: A Comprehensive Survey of Applications, Challenges, Datasets, Limitations, and Future Prospects

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**Abstract**

Within the vast expanse of computerized language processing, a revolutionary entity known as Large Language Models (LLMs) has emerged, wielding immense power in its capacity to comprehend intricate linguistic patterns and conjure coherent and contextually fitting responses. LLMs are a type of artificial intelligence (AI) that have emerged as powerful tools for a wide range of tasks, including natural language processing (NLP), machine translation, vision applications, and question-answering. This survey provides a comprehensive overview of LLMs, including their history, architecture, training methods, applications, and challenges. We begin by discussing the fundamental concepts of generative AI and the architecture of generative pre-trained transformers (GPT). We then provide an overview of the history of LLMs, their evolution over time, and the different training methods that have been used to train them. We then discuss the wide range of tasks where they are used and also discuss applications of LLMs in different domains, including medicine, education, finance, engineering, media, entertainment, politics, and law. We also discuss how LLMs are shaping the future of AI and their increasing role in scientific discovery, and how they can be used to solve real-world problems. Next, we explore the challenges associated with deploying LLMs in real-world scenarios, including ethical considerations, model biases, interpretability, and computational resource requirements. This survey also highlights techniques for enhancing the robustness and controllability of LLMs and addressing bias, fairness, and quality issues in Generative AI. Finally, we conclude by highlighting the future of LLM research and the challenges that need to be addressed in order to make this technology more reliable and useful. This survey is intended to provide researchers, practitioners, and enthusiasts with a comprehensive understanding of LLMs, their evolution, applications, and challenges. By consolidating the state-of-the-art knowledge in the field, this article is anticipated to serve as a valuable resource for learning the current state-of-the-art as well as further advancements in the development and utilization of LLMs for a wide range of real-world applications. The GitHub repo for this project is available at Github-Repo.

**Index Terms**

Large Language Models, Large Vision Models, Generative AI, Conversational AI, LangChain, Natural language processing, Computer Vision, GPT, ChatGPT, Bard, AI chatbots

# LLMs: A Comprehensive Survey of Applications, Challenges, Datasets, Limitations, and Future Prospects

## I. Introduction

Language modeling (LM) is a fundamental task in natural language processing (NLP) that aims to predict the next word or a character in a given sequence of text [1], [2]. It involves developing algorithms and models that can understand and generate coherent human language. The primary objective of LM is to capture the probability distribution of words in a language, which allows the model to generate new text [3], complete sentences [4], and predict the likelihood of different word sequences [5], [6]. They are broadly categorized into statistical language models, machine learning models, deep learning models, and transformer based models as shown in Fig. 1. Early language models, such as n-gram models [7], were based on simple statistical techniques that estimated the probabilities of word sequences using frequency counts [8], [9]. However, with the rise of deep learning in NLP [10], the availability of enormous amounts of public datasets [11], and powerful computing devices [12] to process these big data with complex algorithms, has led to the development of large language models.

Large Language Models (LLMs) [13], sometimes referred to as "transformative [14]" or "next-generation [15]" language models, represent a significant breakthrough in NLP [16]. These models leverage deep learning techniques, particularly transformer architectures [17], to learn and understand the complex patterns and structures present in language data [18]. A key characteristic of LLMs is their ability to process vast amounts of data, including unstructured text, and capture semantic relationships between words and phrases [19]. These models can also process visual [20], audio [21], audiovisual [22], as well as multi-modal data [23] and learn the semantic relationships between them. These models have significantly enhanced the capabilities of machines to understand and generate human-like language [24].

The history of LLMs can be traced back to the early development of language models and neural networks [25]. The journey begins with the era of statistical language models [26]. In this stage, researchers primarily relied on probabilistic approaches [27] to predict word sequences. Classic examples include n-grams, Hidden Markov Models (HMMs) [28] and Maximum Entropy Models [29]. N-grams, for instance, are sequences of adjacent words or tokens that are used to predict the likelihood of the next word based on the preceding ones [30]. While rudimentary by today's standards, these models marked a crucial starting point in the field of natural language understanding. They allowed for basic text generation and word prediction but were limited in their ability to capture complex contextual relationships [31] [32]. [33]. Then a shift towards more data-driven methodologies has been witnessed [34]. Researchers began to explore machine learning algorithms to improve language understanding [35] These models learned patterns and relationships within large text corpora. Support Vector Machines (SVMs) is a notable example from this [36]. Machine learning models brought a more sophisticated approach to NLP tasks, allowing for the development of applications like spam detection [37] and sentiment analysis [38]. Moreover the availability of large-scale Twitter[1] datasets has brought a revolution in real time sentiment analysis [39].

The emergence of deep learning marked a pivotal moment in the development of LLMs [40]. Neural networks, particularly Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks gained prominence [41]. These deep learning architectures delved deeper into the data, allowing them to capture more intricate features and long-range dependencies within text. This stage significantly improved the models' ability to understand context, making them suitable for tasks like machine translation and speech recognition [16], [42]. However, deep learning also faced challenges with vanishing gradients [43] and long-term dependencies [44], limiting their effectiveness.

The breakthrough in LLMs came with the introduction of the Transformer architecture in the seminal work "Attention is All You Need" by Vaswani et al. in 2017 [45]. The Transformer model, based on the self-attention mechanism [46], enabled parallelization and efficient handling of long-range dependencies. It laid the foundation for models like OpenAI's GPT (Generative Pre-trained Transformer) series [47] and BERT (Bidirectional Encoder Representations from Transformers) [48] by Google, which achieved groundbreaking results in a wide range of language tasks. These mechanisms enabled models to consider the entire context of a sentence or document, allowing for true contextual understanding [49]. Transformer-based models, often pre-trained on massive text corpora, can generate coherent and contextually relevant text, revolutionizing applications like chatbots [50], text summarization [51], and language translation [52].

ChatGPT, Llama, and Falcon are all remarkable variants of the GPT (Generative Pre-trained Transformer) model [53], which is developed and pioneered by OpenAI. These models represent OpenAI's ongoing efforts to push the boundaries of

[1]X, as a microblogging platform, allows users to express their thoughts and opinions in short, concise messages called tweets. These tweets often contain rich, real-time information about various topics, making Twitter an excellent source for sentiment analysis

natural language processing and understanding. They share a common foundation in their training methodology, which involves pre-training on vast corpora of text data followed by fine-tuning for specific tasks. During pre-training, the models are exposed to diverse internet text to learn grammar, facts, reasoning abilities, and some degree of common-sense knowledge [54], [55]. This process equips them with a broad understanding of language. Subsequently, fine-tuning is carried out on narrower datasets to specialize the models for particular applications. ChatGPT, for instance, is fine-tuned for conversational contexts, making it suitable for chatbots and virtual assistants [56], [57], [58]. Llama and Falcon, though not as widely known, represent potential advancements or specialized versions, possibly tailored for specific use cases or research objectives. These models collectively exemplify the cutting-edge advancements in natural language processing, enabling more human-like interactions and understanding through the power of AI-driven language models [59], [60], [61].

The training process for models like ChatGPT, Llama, and Falcon consists of several key stages [62], [53]. It begins with a phase known as pre-training, where these models are exposed to a vast and varied dataset of internet text, enabling them to learn grammar, vocabulary, world knowledge, and context [63]. The underlying architecture, based on the Transformer model, is crucial for understanding the relationships between words in sentences. Following pre-training, the models undergo fine-tuning on specific datasets tailored to particular tasks, such as text generation or conversation in the case of ChatGPT. Fine-tuning refines their capabilities for these specialized tasks, with hyperparameter tuning to optimize performance. Ethical considerations are an integral part of the process, aiming to minimize harmful or biased outputs. It's important to note that this training is an iterative and resource-intensive endeavor, continually improved and monitored to enhance both performance and safety [64], [65].

LLMs have undergone several developmental stages, with models increasing in size and complexity. The GPT series, starting with GPT-1 and continuing with GPT-2 and GPT-3 [66], has successively grown in the number of parameters, starting from hundreds of millions (GPT-1) to 1.7 Trillion (GPT-4) [67], allowing for more sophisticated language understanding and generation capabilities [68]. Similarly, BERT-inspired models have seen advancements in pre-training strategies, such as ALBERT [69] (A Lite BERT) and RoBERTa [70], which further improved performance and efficiency.

Furthermore, advancements in LLMs have extended to more specific domains, with models designed for specialized tasks like medical language processing [71], scientific research [72], website development [73] and code generation [74]. Moreover, efforts have been made to address ethical concerns [75], interpretability [76], and reducing biases in LLMs to ensure responsible and equitable use [77]. The development stages of large models have witnessed a constant quest for larger models, improved pre-training strategies, and specialized domain adaptations [78], [79]. As research continues, the potential applications and impact of LLMs on various fields, including education, healthcare, and human-computer interaction, continue to expand, inspiring further innovations and advancements.

In summary and as can be seen from Fig 1; LM research has received widespread attention and has undergone four significant development stages including: statistical language models, machine learning models, deep learning models and transformer-based models[2]. In this research, we mainly focus on LLMs and foundation AI models for language and vision tasks.

Modern language model called ChatGPT [80] was developed by OpenAI [81] and launched in 2022. It is based on the GPT-3.5 architecture [82] and was trained using a sizable amount of internet-sourced text data, including books, articles, wikis and websites (Table I) [83]. ChatGPT is exceptional at producing human-like responses and having conversations with users.

In computer vision (CV), researchers are also actively engaged in the development of vision-language models inspired by the capabilities of ChatGPT. These models are specifically designed to enhance multimodal dialogues, where both visual and textual information are important [84]. Moreover, the advancements in the field have led to the introduction of GPT-4 [82], which has further expanded the capabilities of language models by seamlessly integrating visual information as part of the input. This integration of visual data empowers the model to effectively understand and generate responses that incorporate both textual and visual cues, enabling more contextually rich and nuanced conversations in multimodal settings.

### A. Survey Motivation

The revolutionary ChatGPT has captivated the attention of the community, sparking a wealth of fascinating reviews and discussions on the advancements of LLMs and artificial intelligence [85], [86], [87], [88], [89], [90], [91]. For example, the role of ChatGPT in education is evaluated in [92], healthcare and medicine in [93], [71], protein sequence modeling in [94] and protein generation in [95]. A survey on generative AI is presented in [96], scientific text modeling in [97] and text generation in [98]. The use of LLM in finance is evaluated in [99], impact on labor market in [100] and supply chain in [101], telecom in [102], on code writing capabilities in [103], deep fakes in [104], legal aspects in [105], AI for drug discovery in [106], clinical prediction with LLM in [107], ML for cancer biomarkers in [108], and integration of biotechnology and AI applications to address global challenges in [109]. The advancements in pre-training, fine-tuning, utilization and capability evaluation of LLMs is presented in [85] and a survey on autonomous agents in [110]. The recent progress in visio-language pre-trained models is discussed in [86] and the knowledge graphs construction and reasoning are explained in [111], selection inference is in [76], and quantum-inspired machine learning in [112]. The survey vision language pre-trained models [86] presents an overview of various techniques for encoding raw images and texts into single-modal embeddings as a fundamental aspect, and also discusses

[2]Due to the dominance of Transformer based models, we consider it a different stage, not as a subset of deep learning

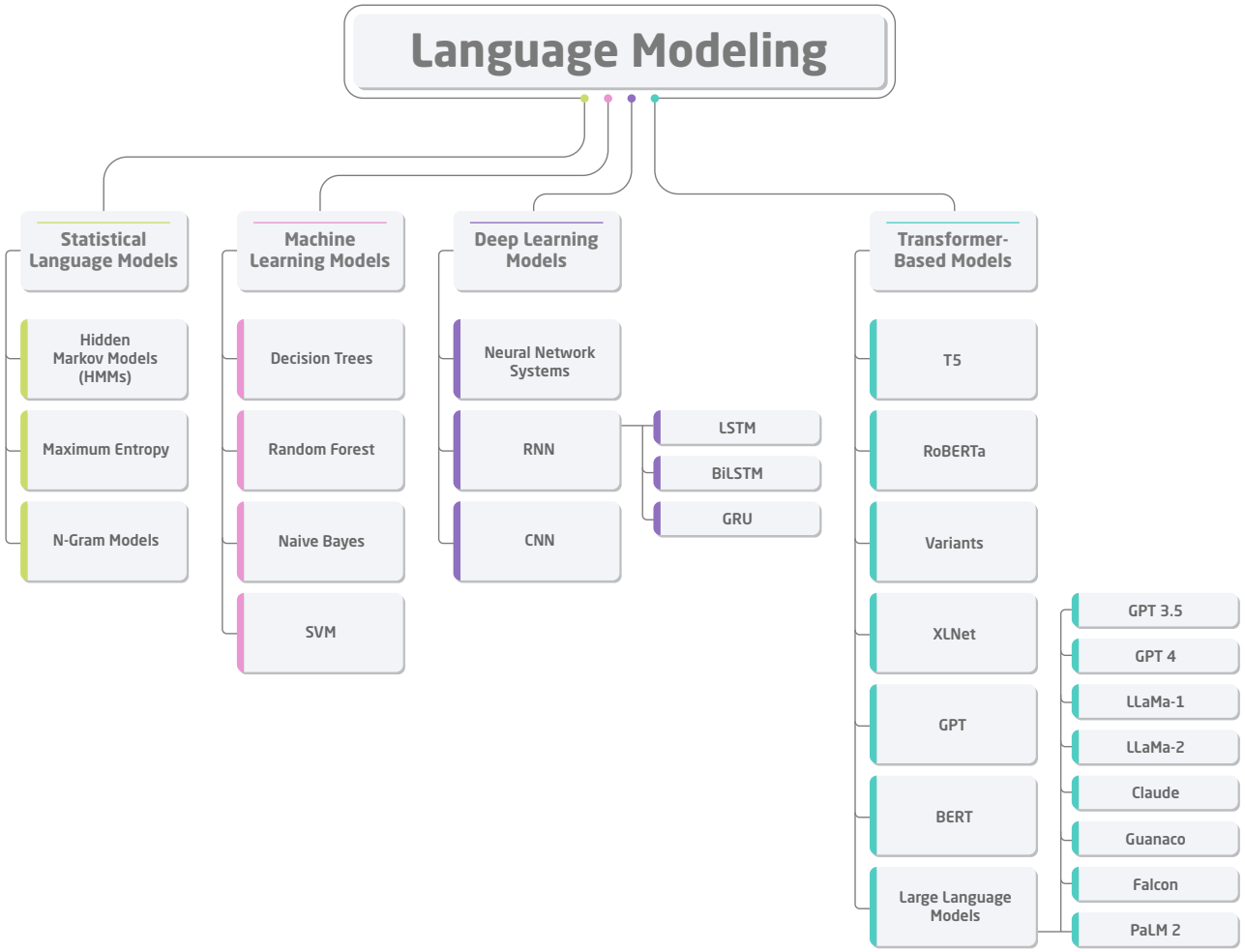



Fig. 1: Types of language modeling. The division of LLMs is categorized into four major blocks: Statistical language models, Machine learning models, Deep learning models and Transformer-based models.

prevalent architectures of vision-language pre-trained models (VL-PTMs), focusing on their ability to effectively model the interaction between text and image representations.

Despite the growing number of studies on LLMs, there remains a scarcity of research focusing on their technical intricacies and effective utilization. Also, the field is progressing at a very fast pace, so a review article with practical applications will contribute a lot to the field. Therefore we also write this paper in the form of an application oriented review. Our primary objective is to explore, learn, and evaluate language models across various domains. We delve into the working principles of language models, analyze different architectures of the GPT family and others, and discuss strategies for their optimal utilization. Furthermore, we provide detailed insights about generative AI, writing prompts, and visual prompting techniques, leveraging GPT-plug-ins, and harnessing other AI/LLM tools. These aspects are generally not covered by the existing related articles. Our comprehensive examination also encompasses a discussion on the limitations associated with the LLMs, including considerations related to security, ethics, economy, and the environment. In addition, we present a set of guidelines to steer future research and development in the effective use of LLMs. We hope that this paper will contribute to a better understanding and utilization of LLMs.

### B. *Contributions*

The main contributions of this article are as follows:

1) Providing a comprehensive overview of GenAI and LLMs, including their technical details, advancements, challenges, capabilities and limitations. We present state of the art analysis and comparisons of different LLMs.
2) Addressing ethical concerns about LLMs, including their computational requirements and potential for perpetuating biases. We also discuss the limitations of LLMs; including, limited understanding of the physical world, tokenization problems, infomration hallucination, fine-tuning and risk of foundation models.
3) Offering insights into the future potential of LLMs and their impact on society and demonstrating the applications of LLM through four practical use cases in the fields of medicine, education, finance, law, politics, media, entertainment, engineering, and others.

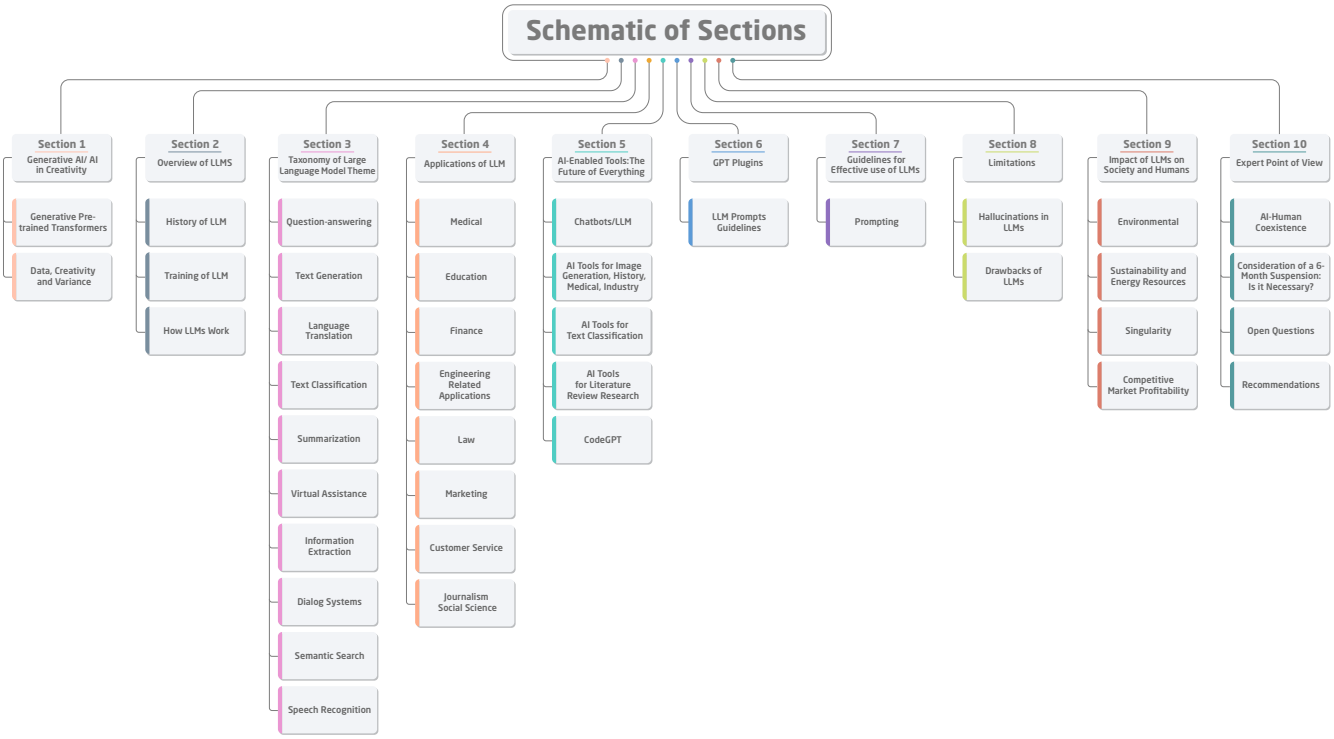



Fig. 2: Schematic Representation of Article Sections - A visual overview of the key sections comprising the structure of the article, providing readers with a roadmap for navigating the content effectively.

TABLE I: **Pre-training data**. Mixtures of data used for pre-training LLaMA [15].

| Dataset | Sampling prop. | Epochs | Disk size |
|---|---|---|---|
| CommonCrawl | 67.0% | 1.10 | 3.3TB |
| C4 | 15.0% | 1.06 | 783GB |
| Github | 4.5% | 0.64 | 328GB |
| Wikipedia | 4.5% | 2.45 | 83GB |
| Books | 4.5% | 2.23 | 85GB |
| ArXiv | 2.5% | 1.06 | 92GB |
| Stock Exchange | 2.0% | 1.03 | 78GB |

4) This article is uniquely presented in a manner to promote practical usage of LLMs, showcasing the actual LLM outputs to corroborate the discussions.

The paper is organized as the following sections. Section II provides an introduction to the role of generative AI. Section III presents an overview of LLMs, summarizing a brief history of LLMs and discussing their training and functionality. Taxonomy of LLMs is presented in Section IV and major applications of LLM through different use cases is discussed in Section V. Section VI explores AI-enabled tools. Section VII discusses the practical use cases of GPT plugins. Section VIII presents guidelines and working examples using prompting techniques. Section IX proposes the limitations and drawbacks of the current state-of-the-art LLM. Section X presents open questions on the subject matter and the authors' perspective on open unanswered avenues. Section XI concludes the survey paper. The overall structure of the article is presented in Fig. 2 for a quick reference at a glance.

## II. Generative AI

Generative AI (GenAI) [113] is perhaps the most disruptive [114] and generalized technology of this decade [115], already influenced many industries, including, Media [116], Marketing [117], Game development and Metaverse [118], Education [119], Software development [120], and Medical [121], construction technology [122], and pharmaceuticals [123]. Unlike general AI systems that perform specific tasks such as data classification [124], clustering [125], object detection [126] and segmentation [127] or predictions [128]; GenAI can generate meaningful new content of multiple data modalities [129]; including, text [3], speech [130], images [131], and videos [132]. Some common examples of GenAI systems are image generators (Midjourney or stable diffusion), Chatbots (ChatGPT, Bard, Palm), code generators (CodeX, Co-Pilot [133]) audio generators(VALL-E)Valle [134], and video generators (Gen-2) [135]

During the past few years, GenAI models size has been scaled from a few million parameters(BERT [48], 110M) to hundreds of billions of parameters (GPT [136], 175B). Generally speaking, as the size of the model (number of parameters) increases, the performance of the model also increases [137], and it can be generalized for a variety of tasks [138], for example, Foundation models [139]. However, smaller models can also be fine-tuned for a more focused task [140].

LLMs, such as ChatGPT by OpenAI, Bard by Google, and Llama by Meta, are a type of GenAI models, specifically designed to generate human-like language in response to a

given prompt [141]. These models are trained on massive amounts of data (see Table I), using techniques to learn the statistical patterns of language. However, many people accord the capabilities provided by GPT models to "more data and computing power" instead of "better ML research" [142].

GenAI works by leveraging complex algorithms and statistical models to generate new content that mimics the patterns and characteristics of the training data [143]. These algorithms may include probabilistic techniques; such as Autoregressive model [144] and Variations Auto-encoders [145], or more recently, Generative Adversarial Networks [146] and Diffusion models [147] or Reinforcement Learning Human Feedback (RLHF) [148].

GenAI has captured significant interest in recent years due to its remarkable performance across an extensive array of applications in text, image, video and generation [149]. Constructed upon the foundation of the transformer architecture [45], these models exhibit an extraordinary capacity to process and generate human-like content by leveraging massive volumes of training data for various topics [150].

### A. Data, Generation, Variance, and Performance measures

To comprehend the intricacies of GenAI systems, it is important to delve into the concepts of data, generation, and variance, and the interplay between them, as they form the foundation of generative systems [151].

*1) Data:* The core of generative AI systems is data. Training models that can successfully capture the underlying patterns and structures of the target domain require high-quality and diverse training data. The generating performance is influenced by the amount, quality, and representation of the training data [152], [153]. Furthermore, the availability of large-scale, labeled datasets allows for the development of more accurate and coherent samples [154], whereas restricted or biased training data may yield sub-optimal results [155].

*2) Generation Process:* GenAI uses the gained knowledge from the training data to generate samples with similar statistical patterns [156]. The generative models are designed to capture the underlying distributions of the training data and generate reliable and realistic samples with properties consistent with the original dataset [157]. The generating process involves approaches; such as adversarial training [158], latent space interpolation [159], and autoregressive modeling [160].

*3) Variance:* Variance is another important factor in defining the diversity and quality of generated samples [161], which shows the variability in the generated samples. A low variance generative AI system may produce similar or repetitive samples, resulting in poor generation, whereas, a high variance, may yield diversified but unrealistic or incoherent samples [162]. Striking a balance between variation and fidelity is difficult in generative AI [163], since it requires managing the trade-off between exploring and exploiting the learnt data distribution [164], [165].

Understanding and regulating the relationship between data, variation, and generation process, is essential for the development of efficient GenAI systems [166]. It entails dealing with issues; including dataset biases [167], mode collapse [168], and balancing exploration and exploitation [**?**]. GenAI systems may generate high-quality, diversified, and realistic samples that correspond with the desired aims and applications by refining the training data, optimizing the generation procedures, and regulating variation [169].[3]

*4) Performance Metrics:* Evaluating the quality and diversity of generated samples is critical for assessing the models' performance [170]. Several strategies for assessing the quality, diversity, and authenticity of generated samples have been established. Here are some common evaluation techniques:

- Visual inspection [171], which is a subjective a evaluation method where human experts or users examine the generated samples and provide qualitative feedback.
- Inception Score (IS) [172] a widely used quantitative evaluation metric for, which measures the quality of generated samples based on their visual appeal and the diversity of the generated classes. Higher IS scores indicate better quality and diversity of the generated samples.
- Frechet Inception Distance (FID) [173] compares the distributions of real and generated samples by calculating the Fréchet distance between their feature representations extracted from a pre-trained Inception model.

Other performance metrics for GenAI systems, include PR-curves [174], coverage metrics [175], user studies and others [176].

### B. Generative AI Design Cycle

A typical design cycle of Generative AI is shown in Figure. 3 as adopted from [129], [177]. GenAI development cycle may be broken into four key steps; (i) define the problem, (ii) model selection or developing from scratch (iii) adapt and align the model, or fine-tuning if necessary, and (iv) finally the deployment and optimization stage [178].

The first stage, scope entails deciding the the nature of the target GenAI model. For e.g., is the target to make it perform well at multiple tasks or only a single task? The nexts stage is the selection of a model. In this stage, a GenAI model developer needs to decide whether to use an existing model for the application or to pre-train one from scratch [179]. In this step, the developer can go for general techniques, for e.g. RNNs, transformers, or pretraining their own model and/or delve deep into creation methods involving more nuanced modifications of the model being used [180].

Following the second stage is an iterative phase of aligning and adapting the model for the scope chosen [181]. This includes steps of prompt engineering [79], which may consist of zero-shot learning, one-shot learning, or a few show learning techniques [182], or even fine-tuning based on the scope [183], and evaluating the model performance.

The last stage is the optimization and deployment for the target application [184]. Since prompting is a fundamental aspect of using and developing GenAI models, we provide a detailed discussion on prompting techniques in this section.

[3]genai meets copyright science paper

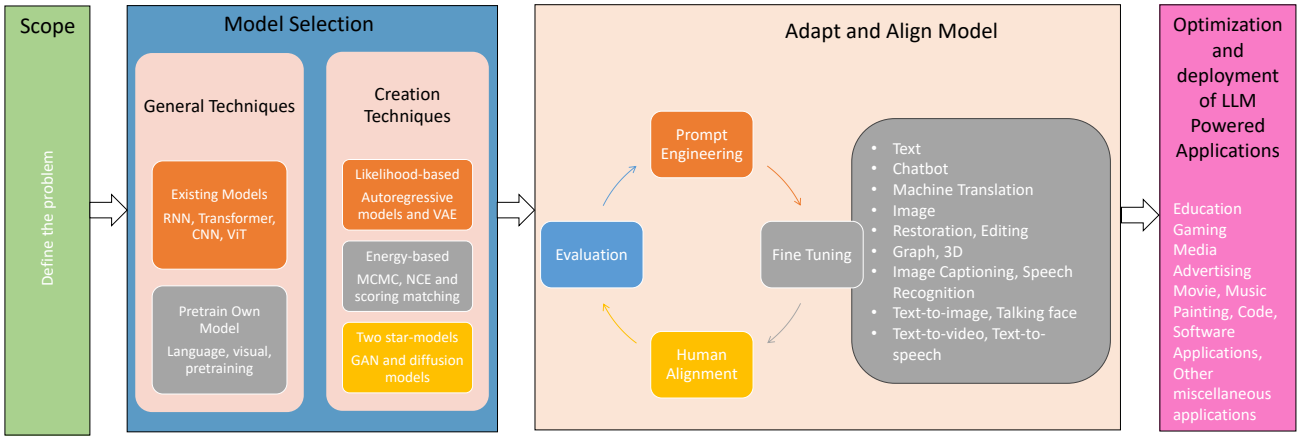



Fig. 3: Generative AI Design Process Schematic. Generative AI models can optimized for a variety of tasks, including, education, healthcare, entertainment and others.

*1) Prompting:* LLMs have given rise to whats called "Prompt Engineering". Prompts are the instructions provided to an LLM to make it follow specified rules, automation of processes and to ensure that the output generated is of a specific quality or quantity [79], [185]. While there is a lack of a formal definition, prompt engineering refers to the designing and wording of prompts given to LLMs so as to get a desired response from them. Writing a prompt appropriately is therefore very important if one needs to use LLMs to assist with tasks in the best manner possible [186].

While some formal techniques such as Explicit instruction (providing a clear direction to the LLM to do something) [187], System Specific Instruction (asking a question from the LLM to answer), Formatting with an example (providing a sample question and its answer and asking the LLM to provide an answer in the same manner), Control tokens (use special keywords in the prompt to help the LLM provide an answer while considering special provided criteria) [188] and Interaction and iteration/chaining (interact with model iteratively to reach to a good answer by fine-tuning on each reply) have been presented [79].

Several different frameworks have been suggested in lieu of prompt patterns for LLMs, these are generic prompt patterns targetting a specific category such as prompt improvement, input semantics etc [79], [189], [190], or prompting for software engineering tasks [191], [192], however, in this work, we aim to present some sets of commands to help users get the most out of the LLMs capabilities from a generic perspective.

- *Defining the role/context*: This should be the first prompt for the LLM. An example of this prompt could be: "Act as a secretary to the Chair of the department", "Act as a Lawyer" or "Act as my programming tutor for Python". By defining a role for the LLM, one can direct it to provide replies or do tasks as a human would do when provided information to work on [193]. A similar first prompt could be providing the context. This can be performed to give the LLM a background of the conditions in which the LLM is supposed to work. For e.g., "We are a company performing mobile application development for Fortune 500 organizations". This can then be followed up with aspects like actions, tasks to perform, steps to follow, etc as mentioned before [194].
- *Prompt creation*: Another interesting prompt command is to ask the model to generate prompts for a certain task [195]. This way, the LLM can be used to generate optimized prompts for tasks that need to be done. An example of this could be: " You are a large language model and are an expert in generating prompts for ChatGPT. Please generate the best prompts on extracting important information from my time series data".
- *Chain of thoughts*: Chain of thoughts prompting [196] in the context of Language Models (LMs) refers to the practice of providing a series of related prompts or partial sentences to guide the generation of coherent and connected text. Instead of providing a single prompt, a chain of thoughts prompt involves providing multiple prompts in succession to encourage the LM to continue generating text that follows a specific line of thinking or narrative [197].
- Other interesting directions in which Prompts can be given are explanation prompts [198] (e.g., "Explain the concept of infinity", Instructional Guides (e.g., "How do I tie my shoe laces"), Extract information (e.g.: one can paste a passage and ask the model to provide answers to questions that one might have), Solve Math problems (e.g., "Find the roots for the quadratic equation, $2x^2 + 3x + 10$ ") and Code help (e.g., "Find the syntax error in the following code") [199].

One concept within prompt engineering is in-context learning [200] in terms of the user "teaching" the LLM to act in a certain manner. The typical prompting scheme in which the LLM is asked to perform a task is an example of zero-shot inference [201], that is, within the context of the current task being worked on, the LLM is asked to perform the task

without providing any sample solution for it. An example of this type prompt could happen in the task of classifying tweets. To perform zero-shot inference, a user will have to just provide the text of the tweet to the LLM and ask it to classify it as positive or negative in sentiment. Another type of prompting could be one-shot inference [202]. In such a case, the user would give an example of a task solution o the LLM and then ask it to perform the task. In the tweet sentiment analysis example previously, this would be the user providing a sample of a tweet and information the LLM that the sentiment is positive and then providing it a second tweet to determine the sentiment of. The third type of prompt is few-shot inference [203], herein, the user provides a few examples of task solutions to teach the LLM about the kind of operation the user wants it to do. For the tweet sentiment analysis example above, it would be providing a tweet/tweets with a positive sentiment and indicating its sentiment and doing so with a negative tweet/tweets as well. Finally, the user can then use the LLM for tweet classification. Using in-context learning allows a user to "fine-tune" the LLM for the specific tasks being performed in the application [204].

*2) Negative Prompting:* Negative prompting [205], [206], [207] provides directions to the LLM about aspects of the prompt that it should avoid generating or deliberately excluding during the generation process [205]. Through the use of negative prompts, one can fine-tune the results generated by the LLM in response to a prompt while being able to keep the prompt generation generic [208]. Another advantage of the use of negative prompting is that it allows for moderation of the output content generated by the model thereby preventing harmful or inappropriate from being generated. "Don't write anything that is offensive or harmful, or factually incorrect." This prompt tells the model to avoid generating text that could be offensive or harmful to others and inaccurate. Notably, the authors in [209] conducted experiments for text based image tranlation and found that negative prompting to be very useful when working with textureless images. Moreover, this type of prompting is very useful when working on text to image generation scenarios and has been incorporated in text to image generation methods such as Muse [210].

*3) Visual Prompting:* Visual prompting [211] refers to the use of visual prompts (such as images or non-visual ones such as music) when providing directions to a model in addition to plain text prompts. The aim is to provide the AI model with a starting point or an example/reference that it can use for the given generative task. For images, this may be given to modify the image provided or generate something that is similar in style, color, or texture etc [212]. This can help in generating content that is closer to a user's expectation from the generative AI being used.

An image-based example of visual prompting could be providing a picture of an office and asking the AI to generate a different theme for it, maybe more nature-centric or in a different color or organizational style [213]. Visual prompting provides greater control of the generated output and therefore results in a more accurate result. It should be noted that visual prompting is not related to images only, this is currently being explored for a host of different applications, including, text generation (generating something based on a sample text so as to copy its style of writing for e.g.) [98], the composition of music (wherein the supplied music piece can be used as a reference for the type of music to compose) [214], game development [215] (where a defined game environment may be provided to the model as a starting point and the model is asked to generate new and unique content) and virtual and augmented reality (wherein a set of augmented/virtual reality environments can be provided to further populate/create current/new environments) [216].

## III. Overview of LLMS

Large Language Models have transformed the way we interact with and process language, opening up new possibilities for natural language understanding, generation, and communication [217]. They continue to evolve, pushing the boundaries of what is possible in the realm of language processing and artificial intelligence [218]. In this Section, we briefly discuss the history, evolution, and training of LLMs.

### A. History of LLM

The history of LLMs can be traced back to the early days of NLP research [219], [220]. The first language models were developed in the 1950s and 1960s [221]. These models were rule-based [222] and relied on hand-crafted linguistic rules and features to process language [223]. They were limited in their capabilities and were not able to handle the complexity of NLP [224].

In the 1980s and 1990s, statistical language models were developed [31]. These models used probabilistic methods to estimate the likelihood of a sequence of words in a given context [225]. They were able to handle larger amounts of data and were more accurate than rule-based models [226]. However, they still had limitations in their ability to understand the semantics and context of language [227].

The next major breakthrough in language modeling came in the mid-2010s with the development of neural language models [228]. These models used deep learning techniques to learn the patterns and structures of language from large amounts of text [10]. The first neural language model was the recurrent neural network language model (RNNLM) [42], which was developed in 2010. RNNLM was able to model the context of words and produce more natural-sounding text than previous models [229]. In 2015, Google introduced the first large-scale neural language model called the Google Neural Machine Translation (GNMT) system [230].

The development of LLMs continued with the introduction of the Transformer model in 2017 [45]. The Transformer was able to learn the longer-term dependencies in language and allowed for parallel training [231] on multiple Graphical Processing Units (GPUs), making it possible to train much larger models [232]. The release of OpenAI's GPT-1 [233] in 2018, marked a significant advance in NLP with its transformer-based architecture. With 117 million parameters, GPT-1 could generate contextually relevant sentences, demonstrating the potential of transformers in revolutionizing NLP tasks [234]. While GPT-1 had its limitations, it set the stage for subsequent,

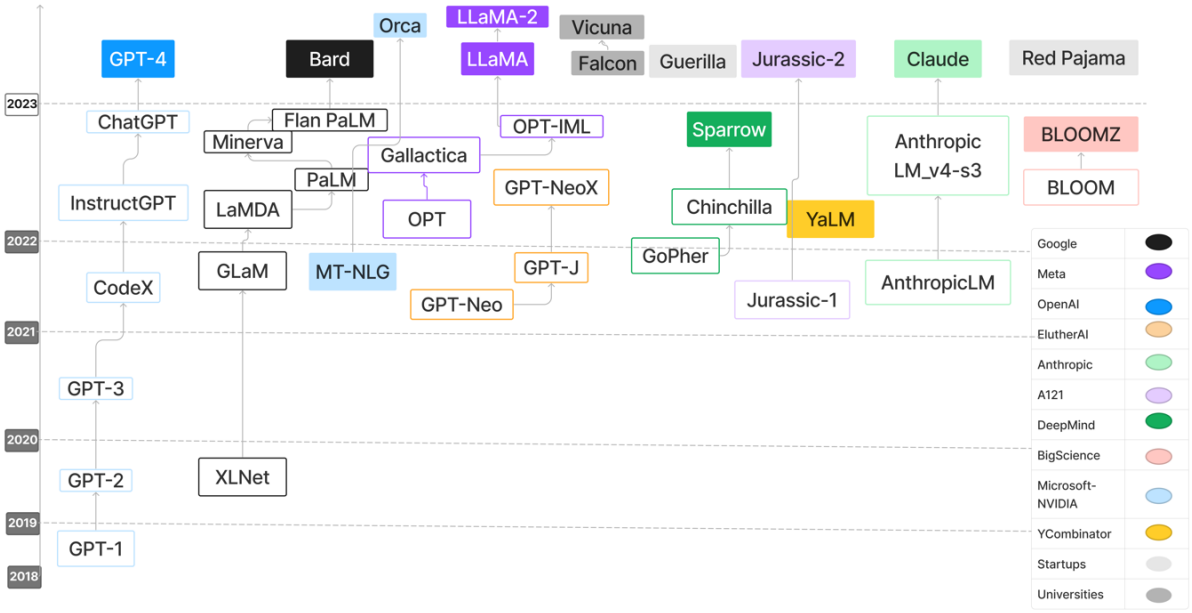



Fig. 4: Illustration of the evolution of Large Language Models (LLMs) over time, highlighting their development across a range of research and commercial organizations. Starting from the initial advancements made in this field, the figure maps out the journey of LLMs, outlining the key milestones, breakthroughs, and model iterations along the way. need to be updated

more powerful models, propelling a new era of AI research and highly-competitive research in LLMs (see Fig. 5).

In 2020, OpenAI released GPT-3 [235], which was able to generate highly coherent and natural-sounding text [236]. GPT-3 demonstrated the potential of LLMs for a wide range of NLP tasks [237]. Inspired by the success of GPT-3, OpenAI released the next iteration of their language model, GPT-4 [238] with the ability to generate even more coherent and natural-sounding text. Following GPT-4's success, Meta also introduced Llama [15], a family of open-source foundation models. Google introduced Bard [239], Amazon introduced AI features in the Alexa [240] models, and Huawei introduced Pangu models [144], joining the AI race.

### *B. Training of LLMs*

Training large language models involves several key steps [242]. The process typically begins with the collection and preprocessing of a large amount of text data from diverse sources [243], such as books, articles, websites, and other textual corpora. The curated dataset [244] serves as the foundation for training the LLMs. After the removal of duplicates [245], noisy and poisonous data [246] and ensuring privacy reduction [247], the training process involves unsupervised learning, where the model learns to predict the next word in a sequence given the preceding context assuming the language generation as a random process [248].

Currently, LLMs utilize Transformers which enable them to model long-range dependencies [249], understand text data [250], enable them to generate new content in the style and characteristics of a genre or author [217]. The training objective is to optimize the model's parameters to maximize the likelihood of generating the correct next word in a given context [85]. This optimization is typically achieved through stochastic gradient descent (SGD) [251] or its variants, combined with backpropagation [252], which computes gradients to update the model's parameters iteratively. Some of the popular transformer-based LLMs are discussed below.

- Bidirectional Encoder Representations from Transformer (BERT): BERT [48] is a prominent language model with significantly advanced NLP tasks. Its training process comprises pretraining and fine-tuning stages [253]. During pretraining, BERT learns a general language representation from large-scale unlabeled text data. It employs masked language modeling (MLM) [254] and next-sentence prediction (NSP) tasks [255].
  MLM involves masking a portion of input tokens and training the model to predict the original masked tokens, fostering bidirectional context understanding [256]. NSP trains BERT to predict whether a second sentence follows the first, enhancing coherence comprehension. After pretraining, BERT undergoes fine-tuning on specific tasks with labeled data. Fine-tuning tailors BERT's learned representations to target tasks, such as sentiment analysis or named entity recognition [257]. Training BERT demands significant computational resources [234], utilizing high-performance hardware like GPUs or Tensor Processing Units (TPUs) or field programmable gate arrays (FPGAs) [258], [259], [260]. Techniques such as layer normalization [261], residual connections [262], and attention mechanisms inherent in the transformer architecture further enhance BERT's capacity to capture intricate dependencies and long-range contextual relationships.
- eXtreme Language understanding Network (XLNet): XLNet [263] is a generalized autoregressive [264] pretraining method that surpasses the limitations of tra-

TABLE II: Comparison of Traditional ML, Deep Learning, and Large Language Models

| Aspect | Traditional ML | Deep Learning | Large Language Models |
|---|---|---|---|
| Training Data Size | Medium | Large | Very Large |
| Feature Engineering | Required | Partial | Not Required |
| Model Complexity | Low | High | Very High |
| Interpretability | High | Low | Low |
| Performance | Moderate | High | Very High |
| Hardware Requirements | Standard | High | Very High |
| Scalability | Limited | Moderate | High |
| Fine-tuning | Not Applicable | Possible | Possible |
| Transfer Learning | Limited | Effective | Effective |
| Contextual Understanding | Limited | Moderate | High |

TABLE III: State-of-the-art for LLM training pipeline [241]. Notations: RM: Reward Modeling, RL: Reinforcement Learning, SFT: Supervised Fine-tuned.

| Stage | Pretraining | Supervised-Finetuning | Reward Modeling | Reinforcement Learning |
|---|---|---|---|---|
| Dataset | Raw Internet I | Demonstration | Comparisons | Prompts |
| Algorithm | Language Modeling | Language Modeling | Binary Classification | Reinforcement Learning |
| Model | Base Model | SFT Model | RM Model | RL Model |
| Resources | 100s of GPUs<br>months of training<br>deployable | 1-100 of GPUs<br>days of training<br>deployable | 1-100 of GPUs<br>days of training<br>not deployable | 1-100 of GPUs<br>days of training<br>deployable |

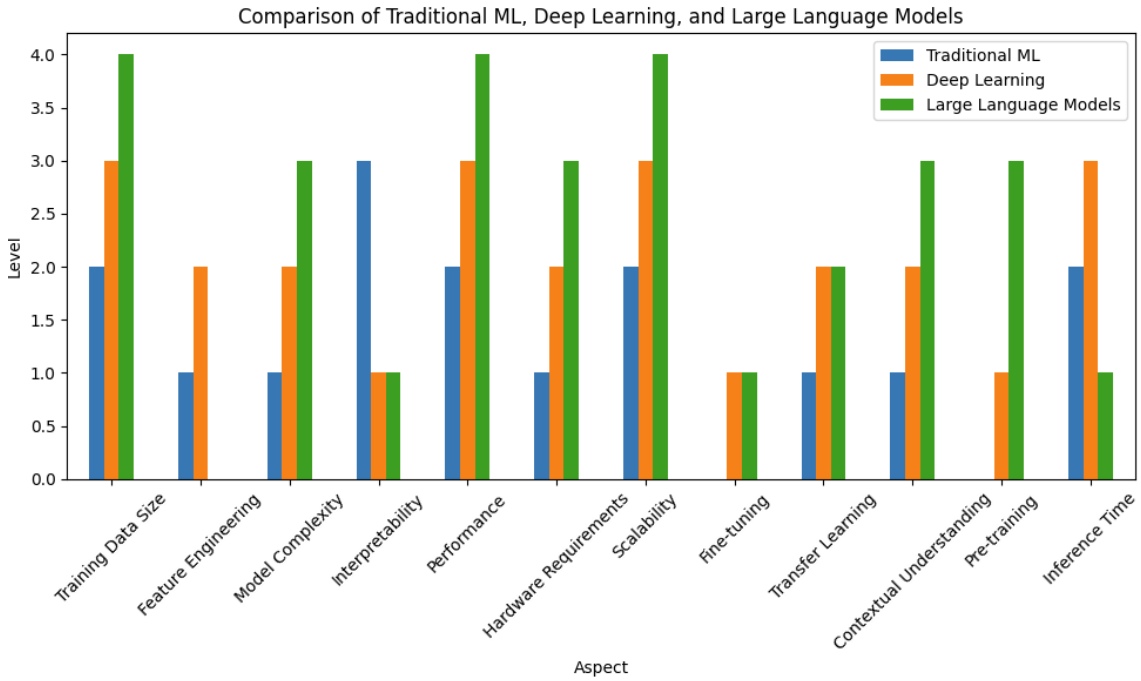



Fig. 5: Comparision of Traditional ML, Deep Learning, and Large Language Models

ditional left-to-right or right-to-left language modeling. XLNet is trained using a permutation-based approach that differs from traditional autoregressive models [265]. The training of XLNet involves two key steps: unsupervised pretraining and supervised fine-tuning. During unsupervised pretraining, XLNet learns to predict words conditioned on the entire input context by maximizing the expected log-likelihood over all possible permutations. This is achieved using a variant of the transformer architecture, similar to models like BERT. The permutation-based objective function used in XLNet training presents unique challenges [266]. Unlike traditional autoregressive models that can rely on the causal order of words for prediction [267], XLNet needs to consider all possible permutations, resulting in an exponentially large number of training instances.

- Text-to-Text Transfer Transformer (T5): T5, developed by Google, is a versatile language model that is trained in a "text-to-text" framework [268]. The key innovation of T5 is the formulation of all tasks as text generation problems. This means that every task, including text classification, summarization, translation, and question answering, is cast into a text-to-text format [269]. For example, instead of training T5 to answer questions

directly, it is trained to generate the complete answer given the question and relevant context. In the pretraining phase, T5 is trained using a variant of the transformer architecture. The pretraining objective is typically based on maximum likelihood estimation, where T5 is trained to predict the target text given the source text. Once pretraining is complete, T5 undergoes fine-tuning on specific downstream tasks [268].

- Conditional Transformer Language Model (CTRL): CTRL is a language model designed to generate text based on specific control codes or prompts [270]. One of the unique aspects of CTRL is its conditioning of control codes or prompts. These control codes guide the model's text generation process, allowing users to specify the desired style, topic, or other characteristics of the generated text. The fine-tuning phase of CTRL is crucial for adapting the model to specific tasks or domains [271].

## IV. Taxonomy of Large Language Model Tasks

LLMs have a wide array of uses for the tasks of processing natural language including but not limited to writing, summarization, translation, retrieving information as shown in Fig. 6. In this section, the various tasks of LLMs towards working with developing such systems have been discussed.

### A. Question-answering

Question-answering (QA) systems [272] allow users to obtain direct answers to questions posed in natural language. LLMs have become a key component in building robust QA systems [273]. LLMs can be effectively pretrained on large text corpora and then fine-tuned on QA labeled datasets [274]. This adapts them to extract or generate answers from passages of text. The broad linguistic knowledge learned during pretraining allows LLMs to understand the semantics of questions and use that to reason about potential answers. Fine-tuning on QA data teaches the models to identify relevant context passages and output the correct response [272]. Key benefits of using LLMs include handling complex questions, synthesizing answers from multiple context documents, and generating clarifying responses when a query is ambiguous. LLM-based QA systems have achieved high accuracy on benchmark datasets [275], surpassing previous state-of-the-art methods. They can be deployed via voice assistants [276], search engines [277], and other interfaces to provide users with quick access to information through natural dialog. Ongoing research is focused on improving reasoning abilities, explainability, and efficiency of LLM question answering.

### B. Text Generation

Text generation is a useful application of large language models, which can automate the process of generating content for various purposes [278], such as articles [279], blogs [280], research papers, social media posts, product descriptions, source codes, emails, and more. With their ability to comprehend and generate natural language, these models can produce high-quality content that is both accurate and coherent [281].

### C. Language Translation

LLMs possess the capability to translate text from one language to another with exceptional accuracy and fluency [282]. This feature is beneficial for a range of users, including language service providers, global companies, and individuals, who can utilize these models for real-time translation, localization, and overcoming language barriers in communication [283]. The impressive accuracy and fluency of these models make them a valuable tool for facilitating effective communication across different languages and cultures [284]. This feature has the potential to enhance global collaboration and increase access to information, making it an important area of research and development in the field of NLP [285].

### D. Text Classification

In addition to their text generation and translation abilities, LLMs are also equipped with exceptional organizational capabilities, such as text classification, analysis, and categorization based on predefined labels or topics [286], [287]. This feature enables the models to effectively manage large volumes of textual data, making them highly valuable for a range of tasks such as sentiment analysis [288], spam detection [289], content moderation [290], and customer feedback analysis [291]. By automating these processes, language models can streamline data management [292], reduce manual labor, and improve the accuracy and efficiency of analysis [293]. These capabilities are particularly useful for businesses and organizations that deal with large amounts of textual data and require effective methods for organizing and analyzing it.

### E. Summarization

LLMs possess the ability to generate concise and coherent summaries of lengthy texts or documents [294]. This feature is highly advantageous for a variety of uses, such as summarizing news articles, research papers, legal documents, and other types of content where extracting essential information is crucial. Summarization by language models can save time and effort while ensuring that the most important points are captured accurately [295]. This feature has the potential to enhance the efficiency and effectiveness of content absorption and subsequent creation, making it a valuable tool for individuals and organizations.

### F. Virtual Assistance

In the realm of virtual assistants and chatbots [296], LLMs play a critical role. These models possess the ability to comprehend user queries, provide relevant information, and engage in natural language conversations. By leveraging large language models, virtual assistants and chatbots can provide highly effective and responsive support to users while also reducing the workload for human operators [297]. This area of research and development is of significant importance, as it has the potential to transform the way users interact with technology and improve the effectiveness and efficiency of customer support and service delivery.

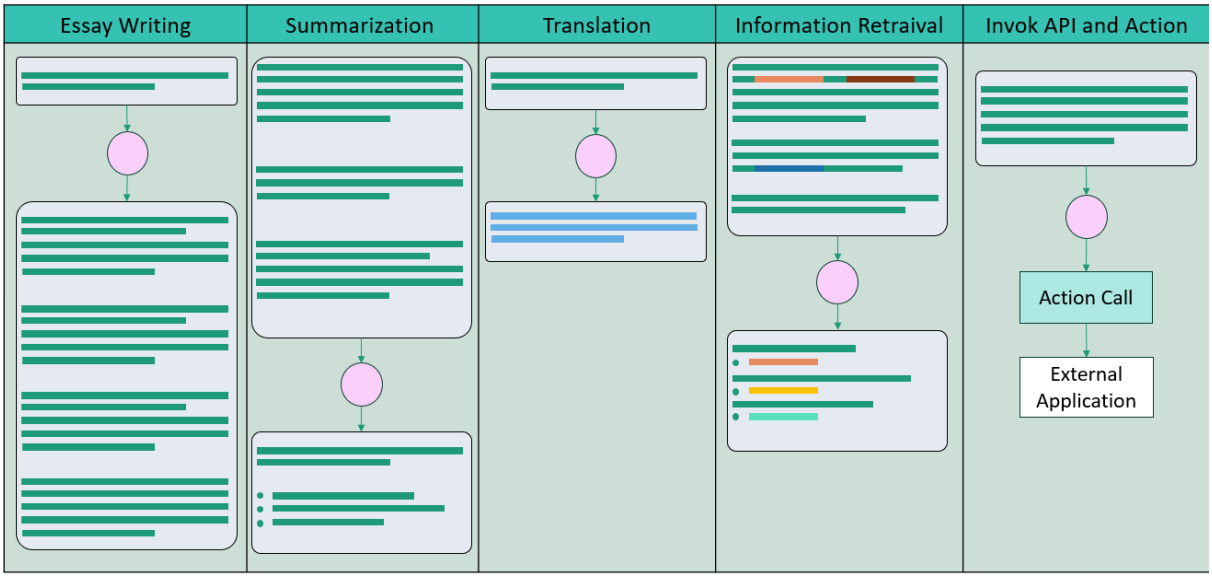



Fig. 6: Examples of LLM. Prompts can be to write an eassay on X topic, summarize a paragrah, translate into X language etc.

### G. Information Extraction (IE)

The use of LLMs in IE is significant for populating knowledge bases. By leveraging fine-tuned LLMs, entities such as people, organizations, and locations, as well as the relationships between them, can be accurately extracted from unstructured text. This process can facilitate the creation of structured knowledge graphs that can be utilized for various purposes [298]. In addition, LLMs assist in event extraction, enabling the identification of key occurrences described in text documents [299]. This feature has the potential to enhance the efficiency and accuracy of information extraction, making it a valuable tool for businesses and organizations that deal with large amounts of textual data. Further research and development in this area can lead to improvements in the quality and effectiveness of IE using LLMs.

### H. Dialog systems

In the context of dialog systems, large language models play a crucial role in facilitating language understanding. The development of large pretrained models like Google's Meena and Microsoft's Blender has led to significant improvements in the naturalness and coherence of open-domain chatbots [300]. These models possess the ability to generate informative, interesting, and harmless responses, making conversational agents much more usable. The application of LLMs in dialog systems has the potential to transform the way users interact with technology, creating more engaging and effective conversational experiences [301]. Further research in this area can lead to improvements in the quality and effectiveness of dialog systems, making them even more valuable for a range of applications and industries.

### I. Semantic Search

In the field of Semantic Search, query understanding is of utmost importance, and LLMs are unparalleled in their ability to discern the underlying intent and meaning of user search queries [302]. This ability enables next-generation search capabilities that go beyond simple keyword matching. For instance, LLMs can recognize that the phrases "best budget laptop"; and "affordable student computer" convey the same information need. This feature has the potential to enhance the accuracy and relevance of search results, making it easier for users to find the information they need. Further research and development in the area of Semantic Search can lead to the creation of more effective and efficient search systems, making LLMs a valuable tool for a range of applications and industries [57].

### J. Speech recognition

Automated speech recognition is a crucial aspect of voice interfaces and transcription. While traditional systems relied on hidden Markov models or Gaussian mixture models, the emergence of deep learning has seen large neural network models like LLMs take center stage in state-of-the-art results. Fine-tuning these models on transcribed speech data using connectionist temporal classification loss enables them to learn acoustic-to-text mappings. This leads to significant improvements in the accuracy of automated speech transcription, even in the presence of accented speech or domain specific vocabulary [303]. The contextual knowledge and continual learning abilities of LLMs make them ideally suited for handling the variability and ambiguity inherent in speech signals. As LLMs continue to increase in scale, they are becoming the standard for building high performance and robust automated speech recognition systems [304].

To sum up the discussion on different LLMs, Table IV provides information on the performance of various LLMs on different reasoning tasks.

TABLE IV: Comparison of LLMs' Reasoning Performance. Notations: MMLU [305]: high school and college knowledge, GSM8K: elementary school math, MATH: very hard math and natural science. All current models struggle, BBH [306]: a collection of 27 hard reasoning problems, HumanEval [307]: a classical dataset for evaluating coding capability, C-Eval [308]: a collection of 52 disciplines of knowledge test in Chinese, TheoremQA [309]: a question-answering dataset driven by STEM theorems. [310], [305], [15], [311], [312], [308]

| Model | Param. | Type | GSM8K | MATH | MMLU | BBH | HumanEval | C-Eval | TheoremQA |
|---|---|---|---|---|---|---|---|---|---|
| GPT-4 | - | RLHF | 92.0 | 42.5 | 86.4 | - | 67.0 | 68.7* | 43.4 |
| claude-v1.3 | - | RLHF | 81.8* | - | 74.8* | 67.3* | - | 54.2* | 24.9 |
| PaLM-2 | - | Base | 80.7 | 34.3 | 78.3 | 78.1 | - | - | 31.8 |
| GPT-3.5-turbo | - | RLHF | 74.9* | - | 67.3* | 70.1* | 48.1 | 54.4* | 30.2 |
| claude-instant | - | RLHF | 70.8* | - | - | 66.9* | - | 45.9* | 23.6 |
| text-davinci-003 | - | RLHF | - | - | 64.6 | 70.7 | - | - | 22.8 |
| code-davinci-002 | - | Base | 66.6 | 19.1 | 64.5 | 73.7 | 47.0 | - | - |
| text-davinci-002 | - | SIFT | 55.4 | - | 60.0 | 67.2 | - | - | 16.6 |
| Minerva | 540B | SIFT | 58.8 | 33.6 | - | - | - | - | - |
| Flan-PaLM | 540B | SIFT | - | - | 70.9 | 66.3 | - | - | - |
| Flan-U-PaLM | 540B | SIFT | - | - | 69.8 | 64.9 | - | - | - |
| PaLM | 540B | Base | 56.9 | 8.8 | 62.9 | 62.0 | 26.2 | - | - |
| LLaMA | 65B | Base | 50.9 | 10.6 | 63.4 | - | 23.7 | 38.8* | - |
| PaLM | 64B | Base | 52.4 | 4.4 | 49.0 | 42.3 | - | - | - |
| LLaMA | 33B | Base | 35.6 | 7.1 | 57.8 | - | 21.7 | - | - |
| InstructCodeT5+ | 16B | SIFT | - | - | - | - | 35.0 | - | 11.6 |
| StarCoder | 15B | Base | 8.4 | 15.1 | 33.9 | - | 33.6 | - | 12.2 |
| Vicuna | 13B | SIFT | - | - | - | - | - | - | 12.9 |
| LLaMA | 13B | Base | 17.8 | 3.9 | 46.9 | - | 15.8 | - | - |
| Flan-T5 | 11B | SIFT | 16.1* | - | 48.6 | 41.4 | - | - | - |
| Alpaca | 7B | SIFT | - | - | - | - | - | - | 13.5 |
| LLaMA | 7B | Base | 11.0 | 2.9 | 35.1 | - | 10.5 | - | - |
| Flan-T5 | 3B | SIFT | 13.5* | - | 45.5 | 35.2 | - | - | - |

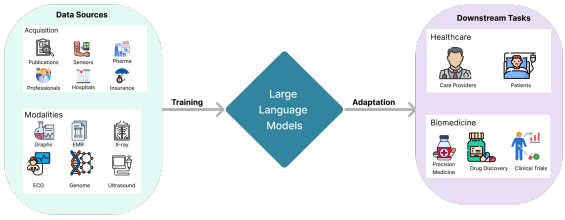



Fig. 7: Interactive framework: LLMs leveraging multimodal healthcare data for diverse tasks in healthcare and biomedicine.

## V. Applications of Large Language Models

Given LLMs wide range of applications, we provide a discussion of their usecases in the fields based on their significance, relevance, and potential impact within their respective domains [260], [313]. An interactive framework for the integration of LLMs in the healthcare ecosystem is shown in Figure 7.

### A. Medical

LLMs like ChatGPT have exhibited remarkable potential in diverse healthcare applications [314], [315], [316]. In medical education, ChatGPT has emerged as an interactive tool that aids learning and problem-solving [317], [318]. Notably, ChatGPT's performance in the United States Medical Licensing Exam (USMLE) was comparable to or exceeded the passing threshold, without requiring specialized training or reinforcement [317]. Moreover, ChatGPT's explanations displayed a high level of concordance and insightful understanding [315].

In [275], introduced MultiMedQA, a new benchmark dataset for evaluating LLMs on clinical tasks. MultiMedQA combines six existing medical question-answering datasets spanning professional medicine, research, and consumer queries. They introduced the concept of instruction prompt tuning [319], which can be used to improve the performance of LLMs on a variety of clinical tasks.

In [320], the potential of ChatGPT in radiologic decision-making is emphasized, showcasing its feasibility and potential benefits in enhancing clinical workflow and promoting responsible utilization of radiology services. Similarly, Kung et al. [315] concluded in their research that LLMs, including ChatGPT, have the capacity to enhance the delivery of individualized, compassionate, and scalable healthcare. These models can assist in medical education and potentially aid in clinical decision-making.

In the domain of clinical genetics, Duong and Solomon [321] found that ChatGPT's performance did not significantly differ from humans when answering genetics-related questions. Furthermore, Fijacko [322] evaluated ChatGPT's accuracy in answering questions related to life support and resuscitation. The findings revealed that ChatGPT demonstrated the ability to provide accurate answers to a majority of the questions on the American Heart Association's Basic Life Support and Advanced Cardiovascular Life Support exams. In neurosurgical research and patient care, LLMs have been investigated for their potential role in various aspects, including gathering patient data, administering surveys or questionnaires, and providing information about care and treatment [323]. Furthermore, AI-powered chatbots hold the potential to enhance patient outcomes by facilitating communication between patients

and healthcare professionals. Leveraging NLP, these chatbots can provide patients with information about their care and treatment in a more accessible manner [324]. There are several tools already in use that allow the system to interact with patients such as Ada Health [325], Babylon Health [326], and Buoy Health [327]. There are tools developed to assist medical practitioners. One such tool is XrayGPT [328], it can be used for automated analysis of X-ray images and have the user/patient ask questions about the analysis. Through the chats, the user can get insight into their condition through an interactive chat dialogue. Another big development is the segment anything (SAM) model by meta, which may be fine-tuned for a variety of medical image tasks [329]. In the drug discovery domain, DrugGPT [330] is developed, which can be used to design potential ligands, targeting specific proteins, using text prompts. The authors in [331] propose generalist medical AI (GMAI). GMAI models are trained on large, diverse datasets of medical data, and they are able to perform a wide range of tasks, such as diagnosis, prognosis, and treatment planning.

### B. Education

AI's impact on education, a widely discussed topic, shows promise in areas like providing student feedback, assisting teacher development, and creating personalized learning experiences [332], [333], [334]. However, effectively implementing tech solutions for inclusive quality education poses significant challenges. General-purpose Large models applicable across tasks and subjects, like foundation models, offer a potential solution [335]. Already, foundation models like MathBERT improve performance in specific educational tasks, such as "knowledge tracing" and the "feedback challenge"

Since the advent of ChatGPT developed by OpenAI, the way students interact with educational materials, assignments and coursework has become different [336] [337] [338]. The accuracy rate for the exams discussed in [336] was below 70 percent indicating its inability to pass the AHA exams. However, this conclusion was drawn due to a design limitation in their study, where they only generated a single response using ChatGPT, introducing bias and severely underestimating ChatGPT's capabilities in this domain. However, the latest study revealed that ChatGPT's accuracy rate increased to 96 and 92.1 percent for the Basic Life Support (BLS) and Advanced Cardiovascular Life Support (ACLS) exams, respectively, allowing ChatGPT to pass both exams with outstanding results [339].

One of the main advantages of using LLM such as ChatGPT in education is that they can help students complete their assignments more efficiently and provide personalized learning experiences [340]. Additionally, AI bots can help automate the grading process, which can reduce the workload for teachers and enable them to provide more detailed feedback to students [341]. Khan Academy, a nonprofit educational organization, has shown interest in utilizing ChatGPT for its business. They have developed an AI chatbot called Khanmigo, which serves as a virtual tutor and classroom assistant [342]. The goal of incorporating ChatGPT into their platform is to enhance tutoring and coaching experiences by providing one-on-one interactions with students. Khan Academy [4] has been an early adapter of GPT-4 based LLMs working as online tutors, becoming the largest case study for the evaluations of LLMs in an educational context in the process.The incorporation of AI in tutoring and teaching proves that it can be a valuable tool in reducing negativity, particularly the perception that its main purpose is for cheating. Undoubtedly, AI technology is still in its nascent phase, yet it shows great potential in supporting students and catering to their individual requirements. [343].

However, there are also some potential drawbacks to using ChatGPT and AI bots in education. One concern is that these technologies may lead to a loss of creativity and critical thinking skills among students [344]. If students rely too heavily on AI bots to complete their assignments and exams, they may not be developing the skills necessary to think critically and solve problems on their own [340].

*1) Learning in the age of AI:* AI chatbots can serve as a valuable tool to aid in various aspects of syllabus preparation [345]. Course objectives can be generated, relevant topics identified, curricula structured, learning resources gathered and reviewed, assessment methods defined, engaging learning activities established, and a well-balanced course schedule created.

*2) Major issues for AI in Education:* One of the major concerns is the utilization of these tools without proper training. It is crucial to address the issue of inadequate training and contextual fine-tuning for LLMs, as their potential utilization without such preparations raises significant concerns [346]. While it is true that LLMs possess the ability to provide answers to a wide range of questions and assist users in generating responses effortlessly, it is essential for students and scientists [347] to receive adequate training specific to their needs in order to fully harness the capabilities of LLMs. Neglecting the necessity for context-specific training and fine-tuning can render these tools less effective and limit their true potential.

Another concern is the use of AI bots that could lead to a decrease in the number of teaching jobs available, which could further widen the gap between those who have access to education and those who do not [348], [349].

### C. Finance

LLMs are making significant advancements in the finance industry [350] with applications ranging from financial NLP tasks [351], risk assessment, algorithmic trading [352], market prediction [353] and financial reporting [354]. LLM's such as BloombergGPT[99], a 50 billion parameter large language model trained on large diversified financial corpus, has revolutionized financial NLP tasks including but not limited to news classification, entity recognition and question answering. By utilizing the huge amount of financial data available, it is able to enhance customer services drastically by efficiently handling customer queries and providing them with excellent financial advisory.

[4]https://blog.khanacademy.org/harnessing-ai-so-that-all-students-benefit-a-nonprofit-approach-for-equal-access/

In addition, LLMs are being used for risk assessment and management, by analyzing past market trends and data, it is able to identify potential risks and provide mitigation steps through different financial algorithms. Financial institutions can use it for better decision making such as credit risk assessment [355], loan approvals and investments [356]. Algorithmic Trading [357] is another application that can leverage LLMs to identify potential opportunities in the trading market by using its predictive and analyzing capabilities.

However, due to the sensitivity of the financial information and privacy concerns, techniques like data encryption, redaction, and data protection policies should be implemented so that these LLMs can be used efficiently in accordance with data protection policies. In this regard, a recent proposition suggested is FinGPT [358] which is an open-source LLM tailored for finance. It is expected that more work will be carried out in this domain.

### *D. Engineering and similar*

LLMs have gained substantial attention across various fields, and their potential applications in engineering domains are increasingly being explored [359]. In software engineering, CodeGPT can be employed to generate code snippets based on natural language descriptions of desired functionality. Additionally, CodeGPT can assist in debugging code by leveraging its language understanding capabilities to identify errors and suggest potential fixes, thereby streamlining the debugging process and reducing development time[360].

The possibility of LLMs utilization to various calculations in mechanical engineering was attempted in [361]. However [361] encountered instances where incorrect procedures, formulas, or results were provided. None of the tasks yielded an exact solution, leading them to discontinue further research.

In manufacturing, Wang et al. [362] conducted an evaluation of LLMs capabilities in supporting design, manufacturing, and engineering tasks. The results indicate that ChatGPT is impressive in providing information, generating coherent and structured content, and proposing initial solutions. Similarly, Badini et al. [363], performed a study in additive manufacturing troubleshooting and evaluated ChatGPT's expertise in technical matters, focusing on the evaluation of printing parameters and bed detachment, warping, and stringing issues for Fused Filament Fabrication (FFF) methods using thermoplastic polyurethane polymer as feedstock material. It was found that ChatGPT provided remarkable accuracy, correctness, and organization in its responses and its approach to problem-solving offered valuable insights in addressing hurdles.

### *E. Media and Entertainment Industry*

The media and entertainment sector is currently undergoing a transformative phase that revolves around data and prioritizes consumer-centric experiences [364]. Companies of all sizes are now striving to introduce groundbreaking innovations that enable personalized, one-to-one interactions on a large scale [365], [366], [367]. LLMs not only enable the creation of original content but also demonstrate a profound grasp of intricate information and the ability to simulate human-like interactions. This includes MediaGPT, a large language model for the Chinese media domain which was presented recently. It can generate high-quality and relevant outputs for various tasks in the Chinese media domain [368]. Similarly, Robertuito [369] was proposed for Spanish social media.

TABLE V: Unveiling the AI Revolution in Entertainment: Real-world Illustrations

| Tools | Function |
|---|---|
| Scriptbook | A cutting-edge AI-powered script analysis tool, is harnessed by film studios to forecast the commercial triumph of a screenplay. |
| Aiva | AIVA (Artificial Intelligence Virtual Artist) represents an AI-driven music composition tool that generates original music tracks tailored to user preferences. |
| LyricFind | LyricFind takes center stage as an AI-powered lyrics search engine, empowering users to find song lyrics using natural language queries. |
| Ziva Dynamics | Ziva Dynamics showcases an AI-powered software tool tailored for creating authentic 3D character models in films and video games. |
| DeepMotion | DeepMotion introduces an AI-powered animation tool capable of producing lifelike 3D animations for video games and films. |
| Speechify | Speechify is one of the most popular and efficient first AI Voice Over generators for using famous singer's voices for singing differents songs. |

Large AI models can also be utilized for generating attractive advertisements and marketing [100], political speeches, slogans and social media posts [370], and promotional videos [117]. Similarly, leading entertainment networks and applications are using LLM based algorithms that can analyze user data to offer personalized recommendations for movies, TV shows, and music. This helps entertainment companies to retain customers and improve their engagement with their content [371]. Moreover, LLMs automate content curation fostering user satisfaction, retention, and monetization. Recently, many companies have developed and offered their services for media and entertainment purposes. One of the prime examples of such services is Dolly, an LLM-trained model developed by databricks Incorporation [372].

The creation of AI-based newscasters [373] is a recent concept that consists of virtual news presenters or anchors that are generated using AI technologies, particularly LLMs [374]. In April 2023, a Kuwaiti media outlet unveiled a virtual news presenter "Fedha" with plans for it to read online bulletins [375]. At the University of Kent's Centre for Journalism, lecturers are grappling with how to prepare the next generation of reporters for the potentially AI-powered newsrooms of the future [376]. AI algorithms have the capability to analyze user data, providing tailored suggestions for movies, TV shows, and music. This enhances customer retention and boosts engagement with entertainment content. Table V presents the recent tools and applications that are transforming the entertainment industry.

### F. Role of LLMs in the Future of Legal Practice

With advancements in AI and the development of tools such as GPT-4, Bard, Gemini, and Bing, it is aimed that these advancements will empower lawyers to enhance legal research, drafting tasks, and decision-making [377]. This has sparked interest among entrepreneurs developing AI tools [378], law firms integrating AI into their workflow, and law professors exploring AI-based techniques for legal aid [379]. A recent example is the Chatlaw [380] model, which is open-source legal language model. A legal informatics approach was introduced in [381] to align AI with human goals and societal values. By incorporating legal knowledge and reasoning into AI systems, the paper contributes to the research agenda of enhancing the integration of AI and law. In [382], the authors propose legal prompt engineering (LPE) as a means to improve LLM performance in legal judgment prediction tasks. The effectiveness of this method has been demonstrated on three multilingual datasets, showcasing the model's capability to handle the intricacies of legal language and reasoning from various sources of information. LLMs' transformative potential in the legal field is evident from their impressive performance in legal exams. GPT-4 scored in the 90th percentile on the Uniform Bar Examination [61], and ChatGPT autonomously passed four law school final exams at a top law school [383]. These achievements showcase the significant impact of AI language models on legal practice. The authors present Chain-of-Thought (CoT) prompts, which aid LLMs in generating coherent and contextually relevant sentences following a logical structure, simulating a lawyer's analytical approach [384]. The study shows that CoT prompts outperform baseline prompts in the COLIEE entailment task using Japanese Civil Code articles. LLMs have also been utilized to explore fiduciary obligations [55].

In a recent working paper by Choi et al., the authors conducted experiments using ChatGPT to generate answers for four authentic exams administered at the University of Minnesota Law School [385]. In summary, the authors concluded that ChatGPT successfully passed all four exams with an overall average grade of C+. This level of performance would grant it credit towards a JD degree, but it would also place the student on academic probation.

Recently in June 2023, in response to fake case citations generated by ChatGPT and submitted in a court filing, a US judge has imposed a fine of $5,000 (3,935) on two lawyers, along with their law firm Levidow & Oberman [386] [387], due to the fake citations generated by ChatGPT. The fictitious legal research was utilized in an aviation injury claim, and the lawyer admitted to inventing six non-existent cases citations. In Texas, a judge now requires attorneys to verify that no part of a filing was composed by generative AI or, if it was, that a human has verified its accuracy [388] [387]. However, not all judges share the same stance on chatbots in legal proceedings. For instance, Judge Juan Manuel Padilla, based in Colombia, acknowledged using ChatGPT's assistance in a case concerning an autistic child [388].

In light of these examples and use cases, LLMs indeed offer numerous benefits, but it is crucial to recognize and comprehend their limitations. They can serve as a valuable tool for initial research, explanations, and improving efficiency in legal practice.

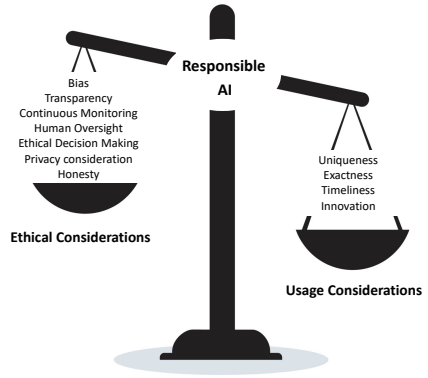



Fig. 8: The Tradeoff for Responsible AI: Example of marketing.

### G. Marketing

Large language models excel in content generation, creating compelling product descriptions, ad copy, blogs, and social media posts, saving time, analyzing SEO optimization, identify keywords, and enhance social media monitoring, vast data, including feedback and social media, offering insights into trends, sentiment, and resonating with audiences [389], [390], [391]. Personalization is a standout feature, allowing marketers to deliver tailored messages based on customer data, improving satisfaction and loyalty [392].

The adoption of LLMs and ChatGPT in marketing offers numerous benefits, but it also comes with potential risks for marketers, consumers, and other stakeholders [391]. The similarity and lack of uniqueness in ChatGPT's responses to similar prompts from different marketers could undermine the distinct identity of the marketer or brand [393] [389] [394]. AI marketing tools like ChatGPT may draw information from unreliable sources, leading to the provision of incorrect information that can lead to false outcomes [391] [395]. Ethics is a significant concern as LLMs can generate content that appears human-generated, raising transparency and disclosure issues [396].

Negative consumer perceptions could arise if AI-generated content is overwhelming or perceived as inauthentic, leading to reduced trust in the brand. Compliance with regulations and data protection laws is crucial to avoid potential legal consequences [397]. Moreover, marketers may become dependent on third-party AI providers, leading to vendor lock-in or reliance on external platforms. To mitigate these risks, marketers must use AI responsibly, provide clear disclosure when AI is involved, and maintain human oversight to verify the accuracy and appropriateness of AI-generated content [398]. It is at the interjunction of the risks (market and consumers both) and ethics where we can balance the scale which opens the window of opportunity for responsible AI adoption (see Figure 8).

*1) Customer Service :* Another application that has witnessed a significant impact is customer service. LLM-powered

chatbots and virtual assistants are increasingly being integrated into customer support systems, providing companies with a scalable and efficient means of addressing customer inquiries and concerns [399], [400], [401]. Unlike human representatives, LLM-powered chatbots can process and respond to inquiries instantaneously, enhancing the overall customer experience [402]. The use of LLMs also leads to cost savings for companies [403]. Due to the high cost associated with employing dedicated customer service agents, an increasing number of companies are exploring the use of NLP to assist human agents [404]. NLP enables the auto-generation of responses that can be directly utilized or modified by agents. In this context, LLMs emerge as a natural and suitable solution.

## VI. AI-Enabled Tools: The Future of Everything

AI tools are becoming increasingly powerful and versatile. They can be used to generate text [405], images [406], and videos [407], translate languages [284], write different kinds of creative content [408], and answer the users questions in an informative way [409]. These powerful tools provide a user-friendly interface for the optimization of daily routine tasks [85]. One such example is the popular website, "There's an AI for THAT" [5], which contains about $7\tilde{K}$ AI tools for $2\tilde{K}$ different tasks. In this Section, we discuss various AI-enabled tools based on LLMs or text prompts.

### A. Chatbots / ChatGPT

Chatbots are frequently used in customer service applications where they can respond to queries, offer assistance, and fix problems [50]. High-tech companies are likely to become even more interested in using chatbots to improve their customer experience and grow their businesses. For example; OpenAI developed ChatGPT [410], Google developed Bard [411], and Meta launched Llama-2 [15]. Here, we critically compare these Chatbots in terms of accuracy, ease of use, cost, integration and others.

*1) Comparison between Chatbots:* ChatGPT and Google Bard are two of the most popular LLMs available today [412], [413]. The third popular LLM being Bing, which is based on GPT-4. Bard is based on the LaMDA [414] (Language Model for Dialogue Applications) architecture, while ChatGPT is based on the GPT-3 (Generative Pre-trained Transformer 3) architecture. ChatGPT was modified and improved using both supervised and reinforcement learning methods [415], with the assistance of human trainers (RLHF) [80]. The learning includes three steps;(i) supervised fine-tuning [416], reward model [417], and maximum policy optimization [418]. First, initiate with pre-trained GPT-3, fine-tune with labeled data. Generate 4-7 responses for input prompts, rank by labelers, creating scalar values for a reward model. Test on new sequences, evaluate responses with reward model, fine-tune parameters for human-like traits through reinforcement learning [419].

LaMDA is a newer architecture (conversational neural language models) that is specifically designed for dialogue applications. Bard uses LaMDA, which is a hybrid architecture that combines batch processing [420] and streaming processing [136]. This allows BARD to handle both historical and real-time data. It is trained on a massive dataset of text and code, while ChatGPT is trained on a massive dataset of text, which means that Bard has a broader understanding of the world and can generate more comprehensive and informative responses, while ChatGPT is better at generating creative and interesting responses.

Both models are capable of generating text, translating languages, writing different kinds of creative content, and answering your questions in an informative way. However, there are some key differences between the two models, such as ChatGPT is more creative, while Google Bard is more authentic. Bard is more personalized than ChatGPT, the responses generated by Bard are more tailored to specific needs, and it is also more scalable than ChatGPT. A comparison between ChatGPT, Bard, and BingChat is made in [411] on VNHSGE [421] dataset, which is a Vietnamese High School Graduation Examination Dataset for Large Language Models.

The results indicate that BingChat performed better than Bard, and ChatGPT. All models perform better than the Vietnamese students [422]. In fact, a comparison between the three popular LLMs services, namely, ChatGPT, Bard, and Microsoft Bing has been of interest to researchers and field practitioners. A recent work by Bhardwaz et. al [423] provided a general comparison of these three models considering accuracy, response time, user experience, and engagement. From their experiments, they found that ChatGPT provided the most relevant responses and accuracy, Bard provided the quickest response and Bing provided the best user experience and engagement. Another comparison by Campello et. al [424] experimented with four different chatbots (above three and Quoras Poe [424]) when asked to solve an intelligence test for recruitment in Brazil found that all four chatbots scored above the 95th percentile while ChatGPT and Bing scored 99th percentile. These are in addition to comparisons being made for typical as well as atypical specific use cases such as news fact-checking (GPT-4 performing the best) [425], taxes [426] as well as political leaning [427] and more. To complete the discussion, Table VIII presents a comparison between ChatGPT, Google Bard, Llama-2, and Microsoft Bing Chatbots.

### B. AI tools for image generation, history, medical, industry

*1) Diffusion Models:* Diffusion models are a scheme of generative models that have provided excellent performance in a variety of applications, most notably the synthesis of images [428]. Starting from a sample of a target data distribution, a diffusion model works in two steps, a forward diffusion process and a reverse diffusion process. The forward diffusion process gradually adds increasing amounts of Gaussian noise[429] to the sample image successively over time, until it becomes the Gaussian distribution [430]. The model is then tasked to start from this noisy image version and undo the noise addition by going through a reverse process to recreate the original data [431], [20].

[5]https://theresanaiforthat.com/

The forward process takes the form of a Markov chain [432] where the distribution at a given time instant only depends on the sample from the timestep immediately preceding it. Therefore, the distribution of the corrupted samples at any given point with respect to the original sample is the product of the successive single-step conditionals up till that point.

Moreover, typically the number of passes of noise addition is in the order of a thousand with the increments each time being quite small. This is necessary to ensure that the reverse process of "recovering" the original sample is more achievable as it has been shown that with infinitely small step sizes, the reverse form will be able to achieve the same functional form as the forward process [433]. Diffusion models use this observation in the forward process.

A diffusion model can be interpreted as a latent variable generative model similar to a variational autoencoder (VAE). The forward process can be thought of as producing latent from data and the reverse process is as converting latent to data. However, as opposed to VAEs, the forward process for diffusion models is typically fixed so that only a single network needs to be trained that deals with the reverse process. The objective function is the variational lower bound on the log-likelihood of the data. It consists of a log-likelihood term or reconstruction term minus a KL divergence term [434] also called the regularization term [435]. The log-likelihood terms [436] encourage the model to maximize the expected density assigned to the data. The KL divergence term encourages the approximated distribution to the prior distribution on the latent variable. Moreover, diffusion models can also be directed to sample conditionally based on a variable of interest which can be incorporated as an additional input during training. This has been the reason that diffusion models have shown better performance than Generative Adversarial Networks [437] in a variety of image generation tasks including perceptual quality ([438], text to image generation ([439], image inpainting [440] and manipulation of images[441].

*2) Image generation:* The images contained in this section were generated by a model incorporating the stable diffusion process in to existing diffusion models as suggested in [442] and uses text to generate photorealistic images. This model was released by stability.ai [443] and was demonstrated to be capable of generating images which were previously difficult to generate, such as images of people with accurate facial features as well as objects with abnormal or impossible shapes. Table VI showcases the output of image generation using various prompts.

In total, nine different prompts were used, these required the AI model to generate humans and natural scenery. The first four prompts tended to depiction of famous personalities (sportsmen and politicians in this case), Muhammad Salah, Lionel Messi, Mike Tyson and Imran Khan. The prompts used were *Mo Salah playing cricket*, *Lionel Messi playing tennis*, *Mike Tyson playing football* and *Imran Khan as a hero*. The second prompt used was regarding the famous painting Monalisa. The prompt was "Generate an image of Monalisa showing her teeth in a wedding ceremony". The third prompt related to natural scenery and was written as *Area of rocks, deep inside the forest, divine domain*. Lastly, the fourth prompt also centered around the generation of humans. In this case, three prompts were given, *A man kissing a girl*, *Generate an image of a guy* and *Generate an image of a woman*.

*3) Video Generation using text prommpts:* Text-to-video generation is a challenging task that involves generating video sequences from textual descriptions or prompts [407]. The video generation process involves text understanding [444], video scene generation [445], temporal structure and transition [446], and video synthesis [447]. One such model is T2V [448], which is a video generation model using text prompts.

### *C. AI tools for text classification*

AI tools are increasingly being used for text classification. Text classification is the process of assigning a category to a piece of text [449]. For example, a text classification tool could be used to classify emails as spam or not spam or to classify news articles as business, sports, or entertainment. Some of the popular libraries include Scikit-learn, NLTK [450], and Spacy [451]. Similarly, Hugging Face's Transformers library [452] is the state-of-the-art toolkit for developers to implement AI text generation capabilities into their applications; including fine-tune models for sentiment analysis, language translation, and text summarization.. This library offers a collection of pre-trained language models, including GPT-3.5 and various other popular models like Bidirectional Encoder Representations from Transformers BERT [69] and RoBERTa [70].

### *D. AI tools for Literature Review Research*

AI tools are increasingly being used to assist with literature review research. These tools can be used to automate tasks such as: Identifying relevant literature, extracting information, and summarizing the content [453], [454]. One such tool is PDFGPT [455], which uses the GPT-3 model to generate responses to user queries.

*1) Fake references:* One of the major drawbacks of using AI tools such as ChatGPT in research is the creation of fake citations and references [456], [457]. Fake citations is an inherent consequence of the generation capabilities of LLMs wherein they may prefer to lean on their generation capabilities rather than on search. When an LLM is asked to provide citations, given its inherent generative nature, it might end up generating text that looks like a research article/source but in reality, might just be a sequence of words that are objectively similar in linguistic/language generation terms [458].

The potential complications that are being created due to the uncontrolled usage of these tools result in many issues among which misleading the scientific community carries vital importance. Fake citations can undermine the credibility of the author [459]. This can lead to inaccurate conclusions and potentially harmful decisions being made based on faulty information. Using fake citations and references can hide the true sources of information used in the research, making it difficult for others to replicate or verify the findings [460]. Recently, WebChatGPT [6] is an impressive extension that has

[6]https://tools.zmo.ai/webChatGPT

TABLE VI: Image generation examples

**Prompt:** Different famous personalities in roles other than their original ones
**Negative Prompt:** blurry, photorealistic

**Generated Images:**



a b c d

**Prompt:** Generate an image of Monalisa showing her teeth in a wedding ceremony
**Negative Prompt:** blurry, low resolution, artistic

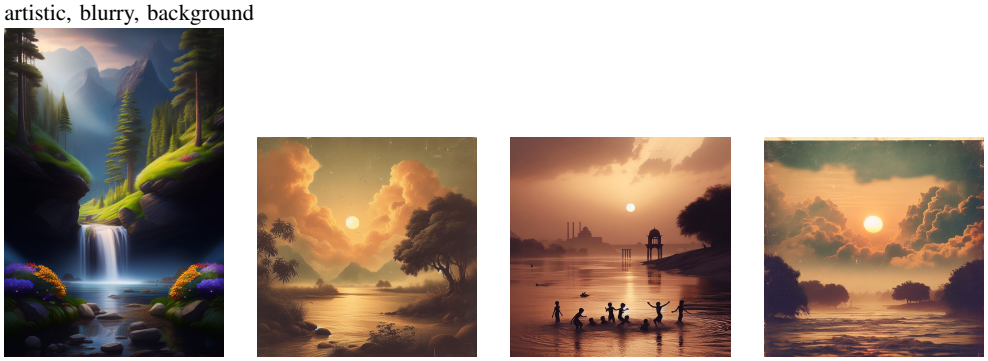
**Generated Images:**



a b c d

**Prompt:** Area of rocks, deep inside the forest, divine domain, river, sunset, kids playing
**Negative Prompt:** artistic, blurry, background

**Generated Images:**



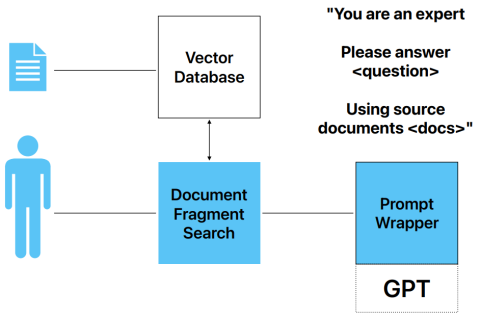
a b c d

**Prompt:** A man kissing a girl/ Generate an image of a guy/ woman
**Negative Prompt:** artistic, blurry, background, young

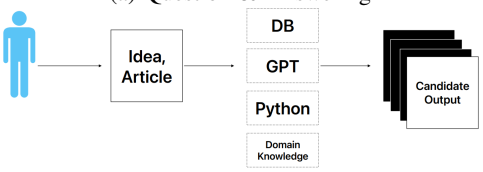
**Generated Images:**



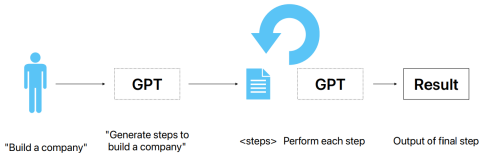
a b c d




(a) Question & Answering




(b) Creative Applications




(c) Multi-step Planning

Fig. 9: Templates for LLM-based application development. GPT is taken as an example scenario representing LLMs.

the potential to address the pervasive issue of fake citations. With the installation of this extension, WebChatGPT becomes equipped with robust capabilities to detect and eliminate fake citations. This advanced tool uses sophisticated algorithms to analyze the authenticity and reliability of citations, ensuring that only accurate and legitimate sources are included.

### *E. AI tools for Coding and Development / CodeGPT*

AI tools are increasingly being used to help programmers write code. These tools can be used to automate tasks such as code completion, refactoring, linting, and testing [461]. GitHub Copilot [462] is an AI-powered code completion tool developed by GitHub in collaboration with OpenAI. It utilizes OpenAI's GPT-3 language model to assist developers in writing code more efficiently. Meta also released the Code-Llama [463], a LLM model, that can use text prompts to generate and discuss code. It has the potential to generate clean and robust code with well documentation in Python, c/C++, Java, PHP, Typescript (Javascript), Bash and other programming languages.

Developers can interact with the LLM by providing prompts or queries related to their coding needs, and the model can generate relevant code segments or suggest solutions to programming problems [464]. LLMs have been used to develop applications in three primary categories which include: (a) Question Answering, (b) Creativity (c) Multi-step planning [464]. These template categories are illustrated in Fig. 9. In Table VII, we present a list of publicly available AI/LLM tools for a variety of applications.

## VII. GPT-Plug-ins

GPT-Plugins are a new way to extend the functionality of ChatGPT. They allow developers to create custom apps that can be integrated into ChatGPT, providing users with new features and capabilities [473]. GPT-Plugins can be used to do things, such as access to external data sources, automate tasks, and enhance user experience [474]. In this Section, we demonstrate several GPT-Plug-ins.

### *A. ChatGPT Plug-ins*

Arguably, the watershed event in the use of ChatGPT was the introduction of plugins by OpenAI. Plugins allow ChatGPT to communicate with third-party sources of data and knowledge bases, thereby providing a platform to extend ChatGPTs capabilities for composition, summarization, nuanced tasks such as sentiment analysis and more to any resource on the internet. Moreover, given that ChatGPT has provided sufficiently acceptable performance for various tasks, plugins allow for ChatGPT to provide answers to queries with updated information from the internet which may not be present in its training dataset [417].

This also has the advantage of providing references for queries to add credibility to answers. For e.g., Bing, the search engine by Microsoft works with OpenAI's ChatGPT through its API to allow its users to ask questions from its Bing search system and get answers with references/sources mentioned. The integration of LLMs in to search engines, thereby allowing users to get answers to human like queries has spearheaded the search engine business in to a new direction [475]. Moreover, this addition of credibility is an important consideration to enable use of ChatGPT and similar LLMs in other critical tasks.

While, at the time of this manuscript, OpenAI still hasn't rolled out plugin development access to all developers, there have been several notable use cases that have already come out. For example, twelve companies have been listed on the OpenAI website [7], namely, Expedia, FiscalNote, Instacart, KAYAK, Klarna, Milo, OpenTable, Shopify, Slack, Speak, Wolfram, and Zapier to have created the first plugins. The power that plugins provide in terms of flexibility to develop new applications has drawn a big attention towards plugin development. Apart from the above-mentioned early developers, two notable plugins already made available by OpenAI are the Code interpreter and the knowledge-based retrieval plugin.

- **Code Interpreter:** The Code interpreter is a built-in Python code interpreter which can be used for performing logical calculations as well as writing code. The interpreter can use the language model's understanding of a human language description of a problem and use that as input to develop Python code for the problem's solution.
- **Knowledge-base retrieval:** A knowledge-based retrieval plugin has also been open-sourced[8] which can be used by developers. This plugin can be used to enable ChatGPT to access data and then use it to gather useful or relevant

[7]https://openai.com/blog/ChatGPT-plugins
[8]https://github.com/openai/ChatGPT-retrieval-plugin

TABLE VII: Publicly available AI /LLM tools

| Tools | Function | Link | Availability |
|---|---|---|---|
| ChatGPT | Conversational AI Chatbot | ChatGPT | Both |
| RoomGPT | Redesign your room in eight different themes | RoomGPT | Public |
| HomeGPT | Redesign your home and office | HomeGPT | Subscription based |
| PDFGPT.IO | Turns PDF into the knowledge base for a ChatGPT type interface | PDFGPT | Subscription based |
| TexGPT | Harnesses GPT-3's power to help you write in Overleaf | TexGPT | Public |
| AcademicGPT | An AI tool to write and review scientific papers, critical analysis and explanation of complex concepts | AcademicGPT | Public |
| DiagramGPT | An AI tool for creating scientific diagrams and flow charts of different processes | DiagramGPT | Public |
| AutoGPT | Auto-prompting without the user intervention | AutoGPT | Public |
| HuggingGPT [465] | A framework to connect various AI models to solve AI tasks | HuggingGPT | Public |
| XrayGPT [466] | Automated analysis of chest radiographs | XrayGPT | Public |
| Video-ChatGPT | A vision language model for video understanding and conservation about videos | Video-ChatGPT | Public |
| ClimateGPT | Large language model for a conversation about the climate in English and Arabic | ClimateGPT | Public |
| CodeGPT | An AI assistant for coding | CodeGPT | Public |
| Code Llama | Open Foundation Models to generate and discuss code | Code Llama | Public |
| MiniGPT-4 [278] | Multi-modal model for a number of tasks, including image generation and website development, using prompts | MiniGPT | Public |
| BiomedGPT [467] | A Unified and Generalist Biomedical Generative Pre-trained Transformer for Vision, Language, and Multi-modal Tasks | BiomedGPT | Public |
| PatientGPT | An AI engine to transform patient navigation with a seamless and customized experience | PatientGPT | Subscription based |
| SentimentGPT [468] | Exploiting GPT for sentiment analysis | SentimentGPT | Public |
| DrugGPT [330] | A GPT based model to design potential ligands, targeting specific proteins | DrugGPT | Public |
| Elicit | AI research assistant, automated literature reviews | Elicit | Public |
| Citation AI | AI research assistant to generate real evidence-based answers | Citation AI | Subscription based |
| Gen-2 | Video generation using text, images, and videos | Gen-2 | Public |
| AI Avatar | Avatar generation | AI Avatar | Public |
| Langchain [469] | Building applications with LLMs through composability | Langchain | Public |

information from the data. These can be files, emails, notes etc. All this by using queries or questions in normal human language.

Lastly, third-party plugins are also an option. These can be created and have been created by several entities. The use of two third-party plugins, namely ShowMe which can be used to generate diagrams and ScholarAI can be used to access academic journals. Table IX provides a list of plugins available for ChatGPT.

## VIII. Guidelines for effective use of Large Language Models

In this section, we provide a list of steps as well as guidelines for effective and responsible use of LLMs [482].

### A. Model selection and deployment guidelines

- **Identify the task**: Determine the task, LLMs can be used for a wide range of tasks, such as text classification, sen-

TABLE VIII: Comparison of Bard, ChatGPT, Llama-2 and Bing Chat

| **Feature** | **ChatGPT** (GPT 3.5) | **Bard** | **Bing Chat** (GPT - 4) | **Llama2** |
|---|---|---|---|---|
| Accuracy | Not as accurate as Bard | Generally more accurate than ChatGPT | Most accurate | least accurate-‘ |
| Versatile | Generally more versatile than Bard | Can generate text, translate languages, and write different kinds of creative content | Not as versatile as ChatGPT or Bard | Less than ChatGPT and Bard both better than Bing |
| Company | OpenAI | Google | Microsoft | Meta |
| Primary Purpose | Creative text generation | Conversational AI | Information retrieval | Text generation, answer questions, language translation etc |
| Integration | Standalone model | Standalone model | Integrated with Bing search engine | Standalone model |
| Easy to use | User-friendly | User friendly | Not as user-friendly as ChatGPT or Bard | User-friendly |
| Access to online data | No, trained on data available till 2021 | Yes | Yes | Yes |
| Cost | GPT 3.5 free / GPT-4 (20 USD per month) | Free | Free | Free |
| Availability | Publicly available | Publicly available | Publicly available | Publicly available |
| Architecture | Generative pre-trained transformer [470] | Pathways Language models (PaLM2) [471] | Next Generation GPT [472] | Transformer |
| Plagriasm detector | Yes | No | No | Less likely to generate plagiarised text |
| Limitations | May generate less coherent or incorrect text | Not as creative as ChatGPT | May provide limited information | Trained on a smaller dataset |

TABLE IX: Some ChatGPT Plugins. This list is not exhaustive and more and more plugins are being developed.

| **Name** | **Task** | **Example use cases** |
|---|---|---|
| Language Translation [284] | Translate between languages | This is particularly useful for documents and information from different languages might need to be translated. |
| Sentiment Analysis [476] | Determine tone of text or conversation | This can be used for the customer analysis and social media monitoring. |
| Spell Checker [477] | Check and correct spelling mistakes | This service can be useful for formal and informal communication such as emails and also browsing the web. |
| Question-Answering [478] | Answer questions for a user query | This can find use in education to build learning platforms, search engines, especially when a more ’understandable’ response is required. |
| Knowledge Graph [479] | Find and present information from a database | Knowledge graphs can be used for improving on search queries and creating recommendations. |
| Speech Recognition [480] | Understand and transcribe speech audio | This service can be used in audio based customer service and also provide services to differently-abled people through audio |
| Emotion Detection [481] | Detect emotion from text or audio | This service can be used for applications relating to market research using verbal ques, used for healthcare as well as assessing reactions to games and other media |

timent analysis, question answering, and text generation [483], [484], [485].

- **Choose the right model**: Choose a pre-trained LLM that is suitable for your task. There are several pre-trained LLMs available, such as GPT-3, BERT, and RoBERTa. Each model has different strengths and weaknesses, so it’s important to choose the one that best fits your needs [250].
- **Fine-tune the model**: Fine-tuning involves adjusting the model’s parameters, such as learning rate, batch size, and number of epochs, to optimize its performance on your task [486].
- **Evaluate the model**: Evaluate the performance of the model on a test dataset. This involves measuring the accuracy, precision, recall, and F1 score of the model on the test dataset [487].
- **Deploy the model**: Deploy the model in your application or system. This involves integrating the model into your application or system and exposing it through an API or user interface. This step also involves setting up monitoring and logging to track the performance of the model in production [488].
- **Monitor and retrain the model**: Monitor the performance of the model in production and retrain it as needed.

This involves regularly checking the performance of the model and identifying any areas for improvement [410].
- **Continuously improve the model**: Continuously improve the model by incorporating user feedback and updating it with new data. This involves collecting feedback from users and incorporating it into the model to improve its performance [489].

### B. Ethical guidelines

The following guidelines will help to ensure the responsible development and use of LLMs focusing on user privacy, bias mitigation, ethical considerations, transparency, competition, collaboration, and environmental impact [80].

- **Protect User Privacy**: LLMs should uphold user privacy and protect user data. This includes safeguarding user-generated content, such as emails, messages, and personal information. Best practices should be followed, such as data minimization, anonymization, and encryption, to ensure user privacy is not compromised [490].
- **Mitigate Bias**: LLMs can inherit and amplify biases present in the data they are trained on. Developers and researchers should actively identify and mitigate bias in their models. This can be achieved through diverse and inclusive training data, bias detection techniques, and evaluation metrics [491].
- **Address Ethical Implications**: LLMs have the potential to be used for harmful purposes, such as spreading disinformation or generating deepfakes [492]. Ethical considerations; including, ensuring accountability, transparency, and responsibility in the development and deployment of models must be taken into account [493].
- **Foster Transparency**: It is crucial that the inner workings of LLMs are transparent and explainable. This can help build user trust and facilitate understanding of the model's behavior. Explainability techniques, such as attention mechanisms and model interpretation tools, can be employed to provide insight into the decision-making process of models [360].
- **Promote Competition**: The development and deployment of LLMs should not be monopolized by a small number of companies or individuals. This can limit innovation and negatively affect competition. Collaboration between academia, industry, and government can foster competition, while also promoting responsible development and use of models [494].
- **Encourage Collaboration**: Collaboration between researchers, developers, and industry should be encouraged to promote the responsible development and use of LLMs. This includes open sourcing models and data, as well as facilitating the sharing of research findings and best practices [495].
- **Minimize Environmental Impact**: Training LLMs can require significant computational resources and energy, which can have negative environmental impacts. Developers should strive to create more energy-efficient models and explore alternative training methods, such as model distillation or transfer learning, to reduce the environmental footprint of models [496], [497].
- **Optimization is exploitation:** is a statement that holds particular significance in the context of LLMs and AI technologies [498]. While these technologies have the potential to revolutionize the way we live and work, they also have the potential to perpetuate existing inequalities and introduce new forms of exploitation [499]. Therefore, it is important to carefully consider the ethical implications of optimization in the development and deployment of LLMs and AI technologies [77].

## IX. Challenges and Limitations of Large Language Models

Although LLMs have made significant contributions to various domains, they have significant limitations and challenges [105], [114]. LLMs are currently perceived as forerunners of Artificial General Intelligence (AGI). However, despite their phenomenal success in conversational tasks, the state-of-the-art LLMs still lack in many aspects that makes them less likely an early manifestation of AGI. We first provide a quick list of the challenges and limitations of LLMs (Fig. 10) and then present a more detailed discussion on a few limitations of critical concerns.

A number of challenges and limitations have been focused on, including biased data, overreliance on surface-level patterns, limited common sense, poor ability to reason and interpret feedback [500], [501]. Other issues include; the need for vast amounts of data and computational resources [502], limited generalizability [503], lack of interpretability [504], difficulty with rare or out-of-vocabulary words, limited understanding of syntax and grammar [505], and limited domain-specific knowledge [506].

The susceptibility to adversarial attacks [242], ethical concerns [75], difficulty with context-dependent language [249], absence of emotion and sentiment analysis [507], limited multilingual capabilities [508], limited memory [509], lack of creativity [408], and restricted real-time capabilities [483] are also critical concerns. High training and maintenance costs, scalability issues, absence of causality, difficulty with multimodal inputs, limited attention span, constrained transfer learning, incomplete world understanding beyond text, insufficient comprehension of human behavior, restricted long-form text generation, limited collaboration, ambiguity handling challenges, inadequate understanding of cultural differences, constrained incremental learning, limitations in handling structured data, and difficulty with noise or errors in input data [510], [511], [512], [513], [514], [515], [92] are some of the key challenges in safe, responsible, and efficient deployment of LLMs.

### A. Training Data Requirements

Large Language Models (LLMs) require a large corpus of data for pre-training the model. Collecting and curating these datasets can be extremely challenging. The size of the dataset makes it impossible to read or assess the quality of the dataset making it prone to having duplicates, making the model biased and degrading its responses [516] [517]. It also makes it difficult to assess the model as the training data may

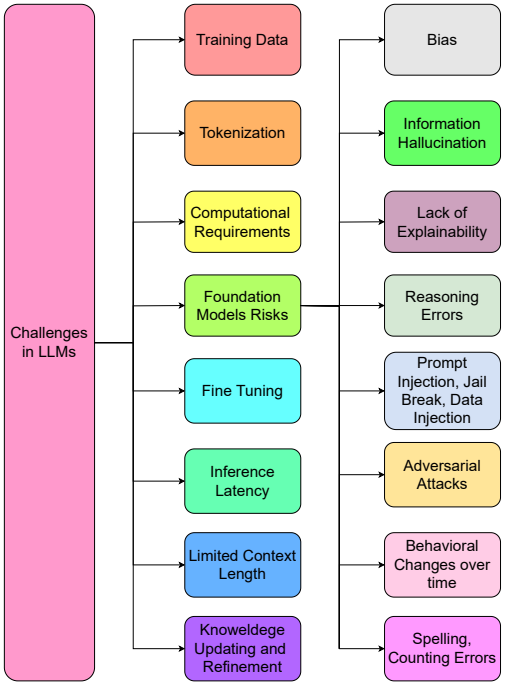



Fig. 10: Challenges in LLMs.

contain data similar to testing samples leading to incorrect evaluation metrics. Since there is no way of checking the datasets manually, it may contain confidential or personal information too, such as telephone numbers leading to privacy leaks during prompting [500]. Due to the fact that the data distribution and requirements in LLMs is more of a black box, it remains uncertain what amount of data is required for different tasks.

### B. Tokenization Problems

LLMs heavily rely on tokenization which consists of breaking down a sequence of words into tokens for the models input. Most LLMs use subward tokenization [56], which is used to create tokens by splitting the words to handle non-familiar vocabulary and at the same time maintaining the computational complexity. However, there are some major drawbacks of tokenization which includes, different combinations of token can be used to relay the same prompts, which may lead to unfair pricing for the APIs of these LLMs. In a multilingual environment it may cause unexpected model responses due to different spacing in the prompts for languages such Taiwanese mandarin or Chinese mandarin [518].

### C. Computational Requirements

Pre-training LLMs requires significant computational costs which can be very expensive, both financially and environmentally. Millions of dollars are spent in training these LLMs with thousands of compute hours and energy consumption. These are classified as Red AI [519], referring to models achieving state of the art results due to vast computation. Scaling these models can also be a challenging task due to the number of resources invested to train these LLMs. The concept of Computer Optimal Training [516] was introduced to address this problem for maximizing the training efficiency with respect to the corpus and model size.

### D. Fine-Tuning LLMs

Fine-Tuning LLMs is a useful technique to train LLMs to custom tasks by training further on these task-specific datasets [48]. However, it requires a high amount of memory and large compute resources to store model gradients, parameters and activations, along with storing these fine-tuned models, limiting its access to a few institutions. Parameter-Efficient Fine-Tuning is a technique that can be used to address this problem which consists of updating a subset of model parameters such as prefix fine-tuning [520], prompt-tuning [521] and adapters [522]. Although techniques like Low-Rank Adaptation (LoRA) [523], or LongLora [524], or QLora [525] can be used to optimize the computation cost, but still computational demands remain a significant barrier for Fine-Tuning LLMs.

### E. Inference Latency

High inference latency is one of the major challenges of LLMs which is mainly due to large memory footprints and lack of model parallelism. Several techniques can be used to mitigate this problem. Efficient Attention [526] can be used for accelerating attention through sub quadratic approximations such as multi-attention query or flash attention. Quantization [59] can be used to reduce the large memory footprint by reducing the computational precision of activations and weights. Pruning [527] and cascading [528] are some more techniques that can reduce the inference latency drastically for efficient and seamless responses.

### F. Limited Context Length

Limited Context Length is one of the crucial aspects of LLMs, as it is extremely useful for interpretation of different prompts and semantic analysis. Without this contextual information, it can drastically degrade the performance of LLMs. There are several strategies that can be used to address this; Positional Embedding Schemas [529], Efficient Attention [530] and Transformer alternatives. Different Positional Embedding Schemes can help LLMs to generalize well to different prompts which may not exist in the training data. Transient Global [531] and Luna [532] are some efficient attention mechanisms that can process larger context lengths effectively. Recurrent Neural Networks (RNNs) [533] and State Space Models (SSMs) [534] are good alternative for transformer-based approaches and are effective for addressing limiting context length.

### G. Knowledge Updating and Refinement

Retraining the models is a costly process and is not sustainable. To address this, approaches such as model editing [535] is a technique which uses non-parametric knowledge resources to alter a model's behavior, and preserving model parameters

by feeding new weights to modify the model's behavior can be used. However, these approaches are found to have limited generalizability and may only be applicable to a limited model architecture. On this end, web plugins and access to the web can alleviate the knowledge updating problem.

### *H. Risks of Foundation models*

A foundation model refers to a base or core model that serves as the fundamental architecture for various machine learning tasks. In [138], a careful assessment of the risks and benefits of foundation models is done. A review [483] also highlights the potential threats and benefits of foundation models in health and education.

- **Bias**: Language models have the potential to unintentionally demonstrate bias when the training data used in their development is biased. According to Schramowski et al. [536], large pre-trained models designed to mimic natural languages can inadvertently perpetuate unfairness and prejudices. The manifestations of these biases are as follows: (i) Training data bias: Language models typically rely on extensive datasets of human language for training [537], [538]. If these datasets contain biases related to factors such as race, gender, or socioeconomic status, the model may internalize and reproduce these biases in its responses. (ii) User interaction bias: The responses generated by Chatbots are influenced by the input received from users. If users consistently pose biased or prejudiced questions, the model may learn and perpetuate these biases in its responses. (iii) Algorithmic bias: Biases can also be introduced through the algorithms employed in training and operating language models and Chatbots [539].
- **Information Hallucination:** Hallucination in Natural Language Generation (NLG) is the generation of text that is nonsensical or unfaithful to the provided source content [540]. Hallucinations in LLMs are often the result of the model's attempt to fill in gaps in knowledge or context, with assumptions that are based on the patterns it has learned during training. This can lead to incorrect or misleading outputs, which can be particularly problematic in sensitive applications [385].
  The cause of hallucinations in LLMs is an area of active research. Recent advances suggest that it's a complex problem related to the model's training process, dataset, and architectural design [541]. In particular, LLMs might be biased towards producing more "interesting" or fluent outputs, leading to a higher risk of hallucination [542]. There have been several proposed methods to mitigate the issue of hallucinations, one approach is to modify the training process to explicitly penalize hallucinations, such as in the case of "reality grounding" [543]. Another is to provide the model with a larger and more diverse dataset, which might reduce the risk of the model making incorrect assumptions [77]. In addition to this, researchers are exploring the use of "verifiable" or "fact-checkable" data during training, to teach the model to rely more on facts and less on its own assumptions [544]. This, however, requires careful consideration of the data and metrics used.
- **Lack of Explainability:** No one can explain a model containing 175 billion parameters, the advent of LLMs has ushered in unprecedented advancements in NLP tasks. However, the sheer complexity and scale of these models present challenges in terms of explainability [545], [546]. As LLMs continue to grow in size, with models containing billions of parameters, the ability to comprehensively explain their decision-making processes becomes increasingly elusive [547], [548].
  This complexity makes it exceedingly difficult for humans to understand and interpret the decision-making mechanisms employed by the model [85]. The lack of transparency [549] hinders the ability to gain insights into how specific inputs lead to particular outputs [550]. This, in addition to the intricate architecture of LLMs, often consisting of deep neural networks, exacerbates the challenge of explainability [551]. The numerous layers and complex interactions make it challenging to trace the reasoning process of the model. While techniques such as attention mechanisms [552] can provide some insights into the model's focus, they do not provide a comprehensive understanding of how the model arrives at its final output.
  Finally, the lack of explainability in LLMs raises concerns regarding accountability, trust, and ethical considerations [553], [549].
- **Reasoning Errors:** LLM can make mistakes in logical reasoning [554], either because of ambiguities in the prompt or inherent limitations in its understanding of complex logical operations. LLMs cannot plan, reason, and have limited knowledge and commonsense [555] about the physical world [556]. From a cognitive science perspective, Auto-regressive LLMs at their best can approximate the Wernicke and Broca areas in the brain [557].
- **Prompt Injection, 'Jail Break' Attacks [558], Data Poisoning Attacks:** GPT-4 is susceptible to various adversarial attacks. For instance, a malicious actor might inject misleading prompts, perform 'jailbreak' attacks to make the model reveal sensitive information, or use data poisoning strategies to manipulate the model's output. Such vulnerabilities have been discussed in [559], [189] through experiments.
- **Adversarial Attacks:** Adversarial attacks on large language models (LLMs) are a type of security threat that can be used to manipulate or control the output of an LLM. These attacks work by deliberately introducing small changes to the input text, which the LLM then misinterprets and produces incorrect or harmful output [560]. One common type of adversarial attack is called a text injection attack. In this type of attack, the attacker introduces carefully crafted text into the input, which the LLM then interprets as a command. For example, the attacker could inject the text "delete all files" into an LLM that is used to control a computer system. The LLM would then delete all of the files on the system [561]. Visual-prompt

based models are also being attacked by these corrupted prompts [562].

- **Behavioral Changes over Time** Chen et. al. [563] investigated the performance of GPT 3.5 and GPT 4 over time, between March 2023 to June 2023, and found that the performance can greatly vary over time. For example, In March, GPT-4 had an accuracy of 84%, but in June, its accuracy dropped to 51%, a decrease of 33%. However, many experts suggest that the performance decrease is due to model drift [564] or prompt drift [565], we need to prompt better for maintaining the performance.
- **Spelling and Counting Errors:** Some specific tasks, like identifying and correcting spelling errors, can be challenging for GPT-4 due to its statistical nature. Another such example are counting errors. Counting error occurs when the model miscounts or misinterprets numerical quantities. For instance, it may provide incorrect calculations or misplace decimal points when performing arithmetic operations, and counting the number of words or characters in long paragraphs [515], [566].

## X. Open Questions

In this section, we evaluate the open questions that are faced by AI researchers from technical, usage and philosophical standpoints.

### A. Environmental and Energy Resources

Studies have revealed that the training process for GPT-3 alone used up 185,000 gallons of water, equivalent to what's needed to fill a cooling tower of a nuclear reactor [567]. This high consumption of water is primarily due to the cooling process of data centers, which necessitates a massive amount of water to regulate the servers' optimal temperature. Moreover, it is expected that the development of newer and advanced version models would need even more significant amounts of water due to their larger data parameters [238]. This concern has been discussed in [568], which presents a method to estimate the water footprint of AI language models and suggests more information transparency in this regard.

Apart from water usage, the training of LLMs demands a considerable amount of electricity. The training of OpenAI's GPT-3 alone resulted in the release of 502 metric tons of carbon, which could provide energy to an average American household for hundreds of years [569]. The amount of energy consumed by AI tools during training can be staggering, with some estimates suggesting that it can take hundreds of thousands or even millions of kWh to train a single large-scale model like GPT-3 [570], [571]. This electricity usage also contributes to indirect water consumption through power generation for data centers located off-site which should be taken into account, leading to carbon emissions [572].

The energy consumption of AI training has significant implications for the environment, particularly in terms of greenhouse gas emissions and climate change [573]. The energy required to train AI models is often generated from fossil fuels, such as coal and natural gas, which emit large amounts of carbon dioxide and other greenhouse gases into the atmosphere. This can contribute to global warming and other environmental impacts [574]. As AI becomes more pervasive in our daily lives, it is important to consider the energy requirements of these systems and develop strategies to mitigate their impact on the environment [575]. One such solution is for data centers to adopt more eco-friendly cooling systems, such as using recycled water or implementing advanced cooling technologies [576]. Additionally, renewable energy sources, such as solar or wind power, can be utilized to power data centers, thereby reducing carbon emissions. Limiting the size and intricacy of LLMs is another potential solution, as smaller models require less data, resulting in reduced energy consumption [572]. Another study by Chien et. al [577] found that with models like ChatGPT, inference services dominated the power consumption and the power emissions for one year were equivalent to 25 times the training power of GPT3. They suggested the use of request direction approaches as a promising manner of reducing power consumption in LLMs.

### B. Ethical Considerations

Inadvertently, LLMs may perpetuate biases inherent in the training data, resulting in outputs that are biased or discriminatory [578] as discussed previously. The challenge lies in identifying and mitigating such biases to ensure fair and equitable treatment across diverse user groups and disciplines [579]. Incorporating robust data authenticity and consent mechanisms, data anonymization techniques, and data retention policies into the development and deployment of LLMs can help ensure the responsible and ethical handling of user data. In [580], presents a comprehensive study of trustworthiness in LLMs, including different dimensions of trustworthiness, established benchmark, evaluation, and analysis.

*1) Humans VS LLMs:* Human interactions offer a deep level of empathy, emotional intelligence, and the ability to understand complex nuances in everyday life-situations. Humans responses are not only based on the current situation (prompt), but also considers other factors [195].

On the other hand, chatbots powered by AI have their advantages. They can operate 24/7, handle large volumes of inquiries simultaneously, and provide quick and consistent responses [581]. There is also a need to develop new performance metrics for measuring the intelligence of AI systems, as traditional methods of assessing intelligence, such as IQ tests [315], are not well-suited for AI systems, as they are designed to measure human intelligence [582].

*2) Copy-right Issues:* Training Large Language Models (LLMs) involves using vast amounts of textual data, which often includes copyrighted material [583]. This practice raises significant legal and ethical concerns regarding the unauthorized use of protected content. [583] examines the AI lifecycle from data collection to model deployment, emphasizing that most training materials are copyrighted, creating legal challenges.

*3) Training data contamination from AI-generated content:* Data sources for training large models are typically scraped from the internet. With the increasing popularity of generative AI, it is possible that data present on the internet will have a

significant component generated by AI models and therefore, reduce the human creativity aspect of the training data. Models, if trained on such data might end up trying to copy the generation aspects of previous AI models rather than humans only. One solution to this could be to use AI detection engines [584] that can determine content generated by AI before passing it through the model during the training process. There is a need to develop a dependable mechanism [491] to perform this task and retain the integrity of data.

*4) The future as we perceive it:* Large Language has vast potential for practical applications, particularly when combined with human oversight and judgement.

- **Use in Low Stakes Applications, Combine with Human Oversight:** LLMs are best suited for low stakes applications, where errors or inaccuracies can be tolerated. Moreover, combining LLMs with human oversight can significantly mitigate the risk of errors, biases, and other issues [585], [586].
- **Source of Inspiration, Suggestions:** LLMs can serve as an invaluable source of inspiration and suggestions, helping users brainstorm ideas [587], create content [588], and make decisions [589].
- **Copilots Over Autonomous Agents:** Given its limitations, LLMs are better suited as a 'copilot' that provides assistance and suggestions, rather than an autonomous agent that acts without human input or oversight [590], [591].
- **Artificial General Intelligence - AGI** Artificial general intelligence (AGI [592]) is a hypothetical type of artificial intelligence that would have the ability to learn and perform any intellectual task. In [310], GPT-4 is found to have sparks of artificial general intelligence. GPT-4 is able to perform a variety of tasks; such as solving math problems, writing creative contents, writing poems and poetry [593] and answering questions in an informative way.

  However, in our opinion, realizing the dream of AGI is still far away, despite of the rapid progress in the LLMs development. The key challenges include; understanding natural intelligence [594], developing adaptable fully autonomous models [595], and being safe and reliable with the understanding of the physical world [596], [597].
- **Embodied Artificial Intelligence** Embodied AI [598] bridges digital intelligence and physical action, enabling systems to perform complex tasks, interact naturally with humans, and adapt to dynamic environments, offering transformative potential across various industries. Vision-language models [599] are critical for enhancing the capabilities of Embodied AI by enabling systems to understand and interpret visual data [600] in conjunction with natural language. One such model is EmbodiedGPT [601], which integrates the powerful capabilities of GPT models with vision-language technology [602].
- **Small Language Models**S mall language models [603]...., Paradigm shift between accuracy and trade-off, small languge models are the future — **need to be extended**
- **Democratizing AI** Democratizing AI [604] is a crucial movement that seeks to make artificial intelligence accessible and inclusive for a wide range of individuals and organizations. By breaking down barriers and providing user-friendly tools, democratization empowers diverse communities to leverage the power of AI to solve problems and drive innovation. It emphasizes the importance of open data, transparency, and accountability, ensuring that AI systems are unbiased, understandable, and ethically grounded.

  *No-code AI* platforms [605] may also assist in democratizing AI initiatives [606], by providing a user-friendly interface that allows users to build and deploy ML models without any coding experience [607]. *No-code AI* can be used to leverage machine learning operations (MLOps) [608], to ensure models are deployed and managed effectively in production.

## XI. Conclusion

In this survey, we provided a comprehensive exploration of LLMs, their implications, technical concepts, and practical learning and usage. We discussed the potential benefits and risks of LLMs, and explored the different ways in which they can be used. We also provided a number of examples of how LLMs are being used in practice; such as generating images, chatting with pdf files, and also discussed GPT plug-ins. A comparision of popular chatbots; such as, ChatGPT, Bard, and Bing Chat is also provided.

We particularly explored the applications of LLMs in medicine, engineering, education, finance, media, law, and the entertainment industry. A list of popular LLM-based open-source applications for a variety of tasks is also presented. By delving into the technical intricacies, effective utilization, and future potential of LLMs, the survey will contribute to a deeper understanding and usage of these models within the research community. The survey has shed light on the key elements that drive the success of large language models through an examination of their working principles, diverse architectures, guidelines for prompting, AI-enabled tools and plug-ins, optimal strategies for employing LLMs, as well as advancements in pre-training, fine-tuning, and capability evaluation. A thorough comparison between popular chatbots has been provided as well.

Furthermore, the survey has also highlighted the importance of the safe and ethical use of AI tools like ChatGPT and others. It recognizes the need for developing guidelines and regulations to address concerns related to security, ethics, the economy, and the environment. Ensuring the responsible integration of LLMs in healthcare, academia, and other industries is critical, as it enables these tools to effectively support and enhance human endeavors while upholding the values of integrity, privacy, and fairness. In our opinion, A technology $X$ can replace a technology $Y$ on a task $Z$, and can also help increase the productivity of humans on several tasks. LLMs have a great potential to transform many fields and bring positive impact on humans and society .

As the field of LLMs continues to evolve and progress, future research and development efforts should focus on

improving the accuracy and performance of these models, addressing their limitations, and exploring new ways to use them. By adopting the guidelines presented in this survey, researchers and practitioners can contribute to the ongoing advancement of LLMs and ensure that they are used in a responsible and beneficial manner.

## Declaration of interest

The authors have no conflicts of interest to declare.

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